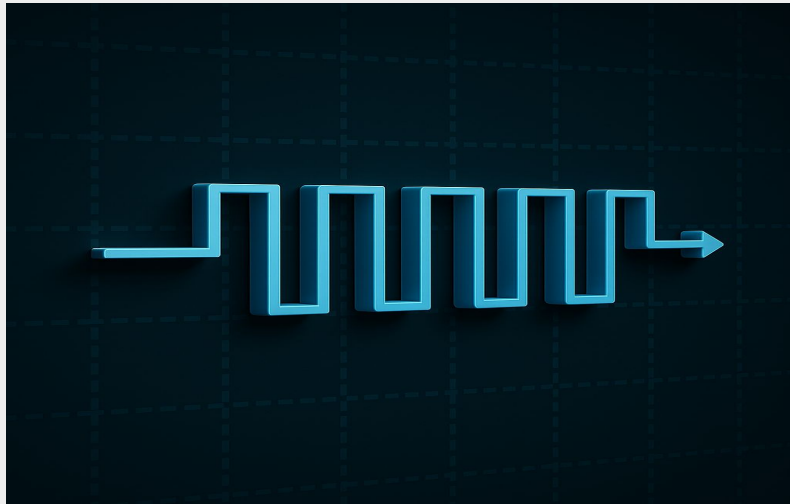
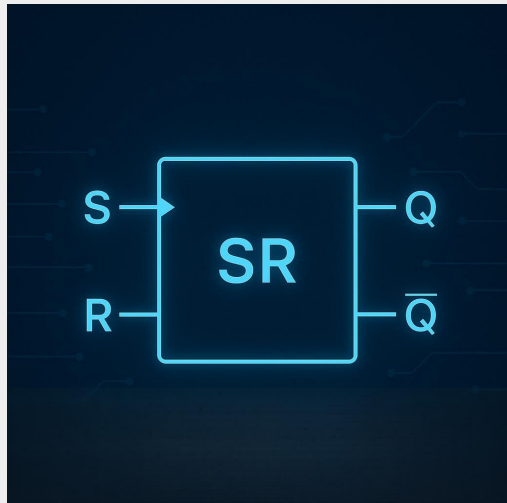


Lecture 8

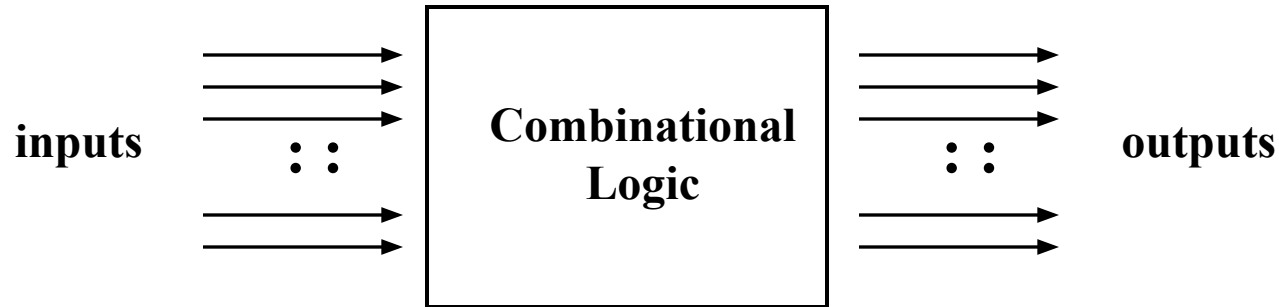
Flip-Flops & Timing Diagrams



CSE260 : Digital Logic Design

Introduction

- Two classes of logic circuits:
 - ❖ combinational
 - ❖ sequential
- Combinational Circuit:**



Combinational Circuit vs. Sequential Circuit

Combinational

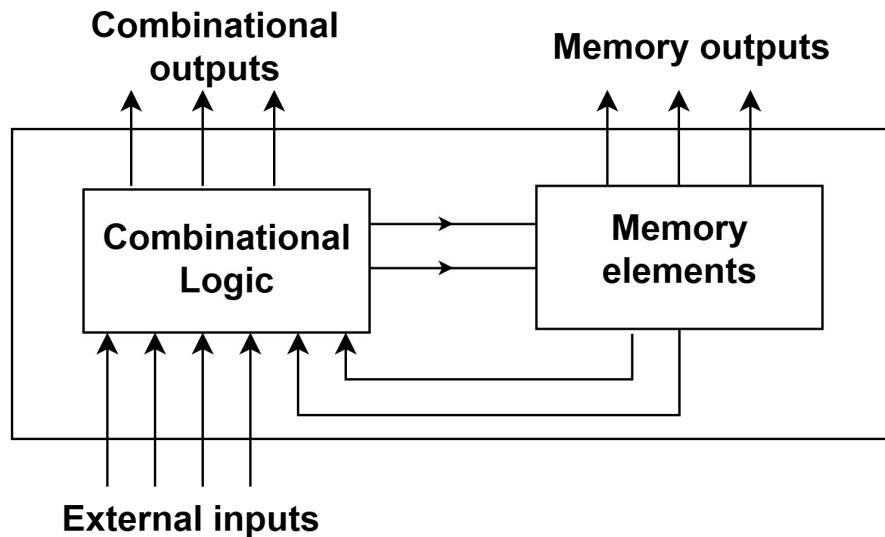
- A combinational circuit consists of logic gates whose **outputs** at any time **are determined directly from the present combination of inputs** without regard to previous inputs.
- A combinational circuit **performs** a specific information-**processing** operation fully **specified** logically **by** a set **of Boolean functions**.

Sequential

- Sequential circuits employ memory elements in addition to logic gates. **Their outputs are a function of the inputs and the state of memory elements**
- The state of memory elements, in turn, is a function of previous inputs. As a consequence, the **outputs** of a sequential circuit **depend not only on present inputs, but also on past inputs**.

Sequential Circuit

- A **sequential circuit** consists of a *feedback path*, and employs some *memory elements*.



Sequential circuit = Combinational logic + Memory Elements

output= external input + present state of memory element



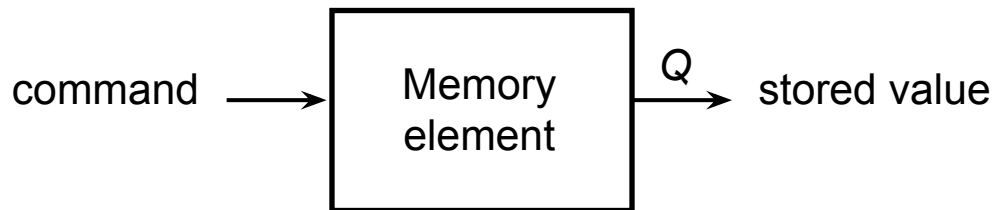
Introduction

- ❖ There are two types of sequential circuits:
 - *synchronous*: outputs change only at specific time (i.e. with clock input)
 - *asynchronous*: outputs change at any time (i.e. without clock input)
- ❖ Multivibrator: a class of sequential circuits. They can be:
 - *bistable* (2 stable states)
 - *monostable* or *one-shot* (1 stable state)
 - *astable* (no stable state)
- ❖ Bistable logic devices: *flip-flops*.
- ❖ Flip-flops differ in the method used for changing their state.



Memory Elements

- ❖ **Memory element:** a device which can remember value indefinitely, or change value on command from its inputs.



$Q(t)$: current state

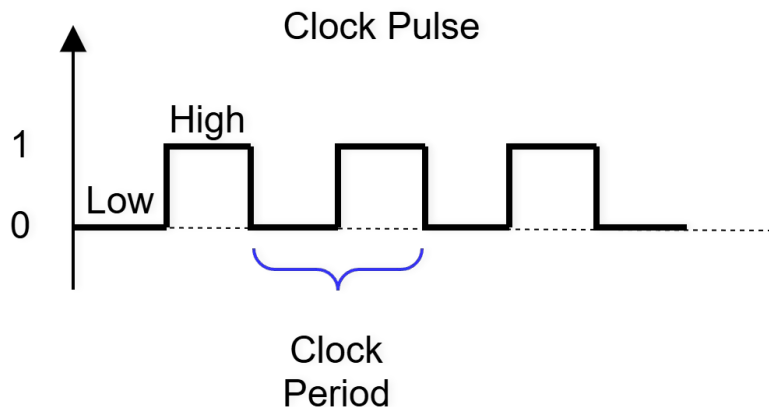
$Q(t+1)$ or Q^+ : next state

Clock Pulse

- A clock pulse is a periodic signal (HIGH/LOW) that synchronizes state changes in sequential digital circuits
- Controls timing in flip-flops, counters, registers

Properties:

- **Clock Period**: The time taken to complete one cycle of a clock signal (measured in seconds (s) or milliseconds (ms)).
- **Clock Frequency**: The number of cycles per second (measured in Hertz (Hz)).



$$Time = \frac{1}{frequency}$$

$$Frequency = \frac{1}{Time}$$

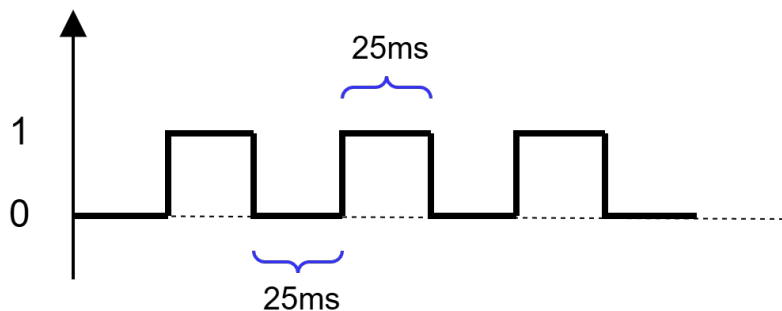
Clock Pulse

Duty Cycle :

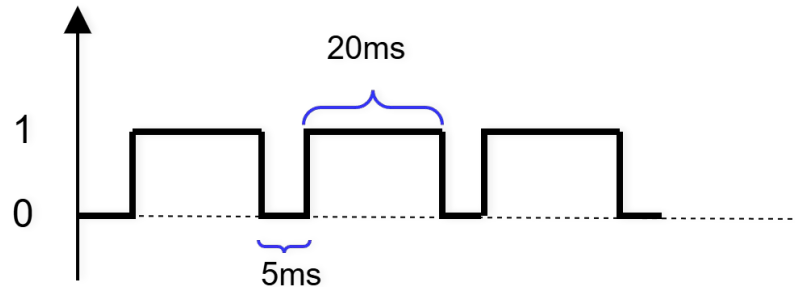
The percentage of time the clock signal stays at a high (1) level within one cycle.

$$\text{Duty Cycle (\%)} = \left(\frac{\text{High Time}}{\text{Clock Period}} \right) \times 100$$

$$\text{Clock Period} = \text{High Time} + \text{Low Time}$$



$$\text{Duty Cycle} = \frac{25}{25 + 25} \times 100 = 50\%$$



$$\text{Duty Cycle} = \frac{20}{20 + 5} \times 100 = 80\%$$

Triggering Mechanisms

- Triggering refers to **how and when** a latch or flip-flop changes state based on the clock signal.
- It determines when the stored value updates in a sequential circuit.

Level Triggering

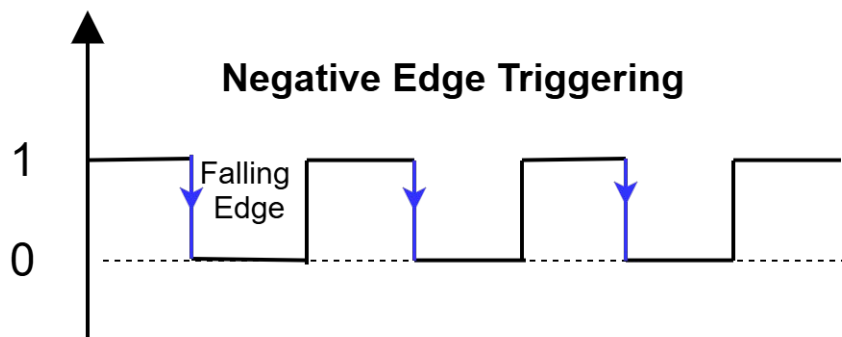
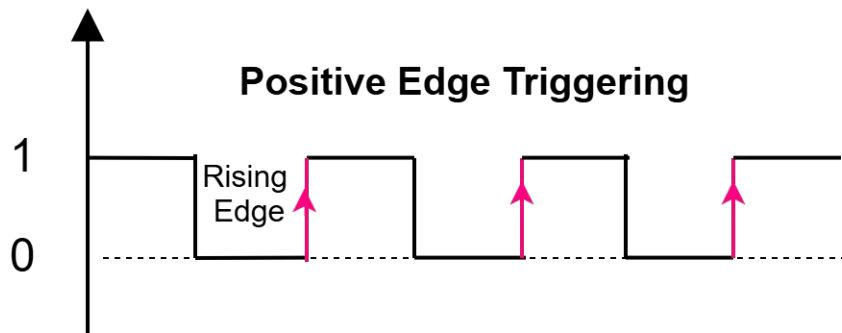
- The circuit gives output during the entire duration when the clock is HIGH.
- **Positive Level**: Active During High
- **Negative Level**: Active During Low
- Mostly Used in latches

Edge Triggering

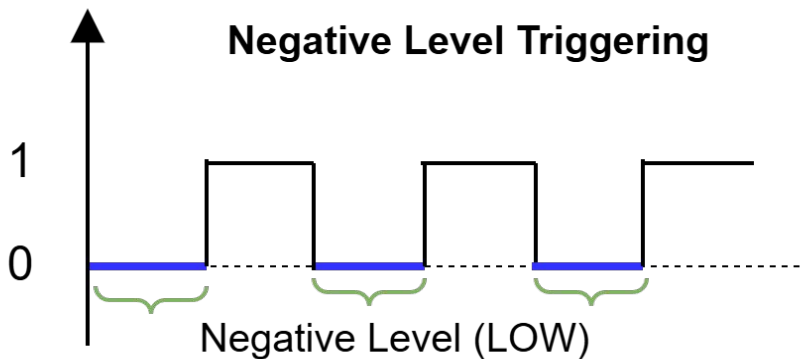
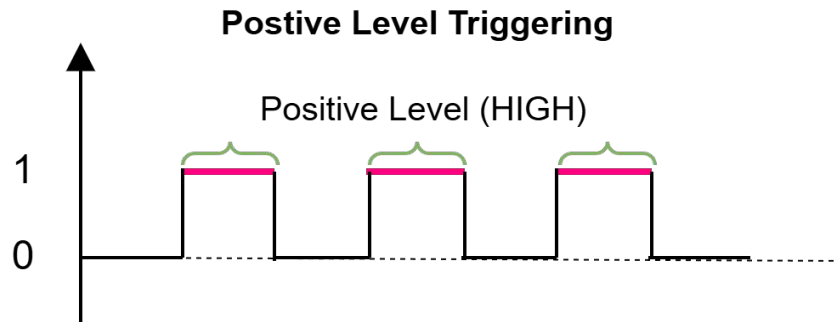
- Circuit reacts only at the transition point of the clock pulse.
- **Positive Edge** ($\uparrow$): Triggers when clock transitions from LOW to HIGH ($0 \rightarrow 1$).
- **Negative Edge** ($\downarrow$): Triggers when clock transitions from HIGH to LOW ($1 \rightarrow 0$).
- Used in Flip flops, Registers

Triggering Mechanisms

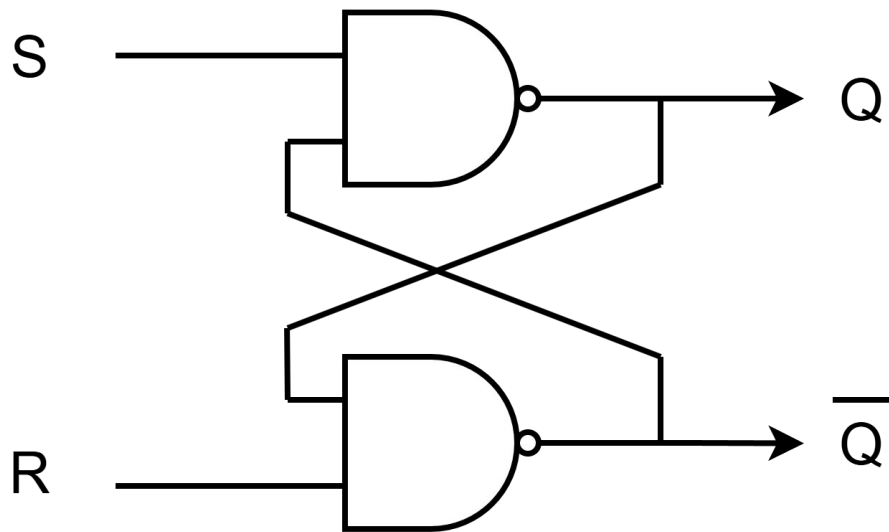
Flip Flop's State Changes



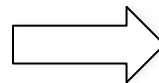
Latch's State Changes



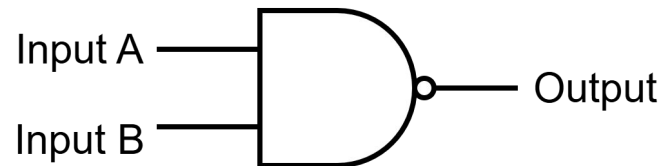
SR Latch Using NAND Gate



We will need the knowledge of this truth table later



2-input NAND gate



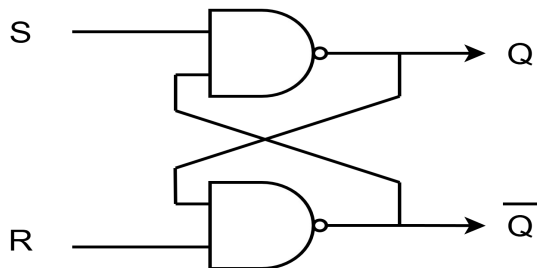
A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

SR Latch Using NAND Gate

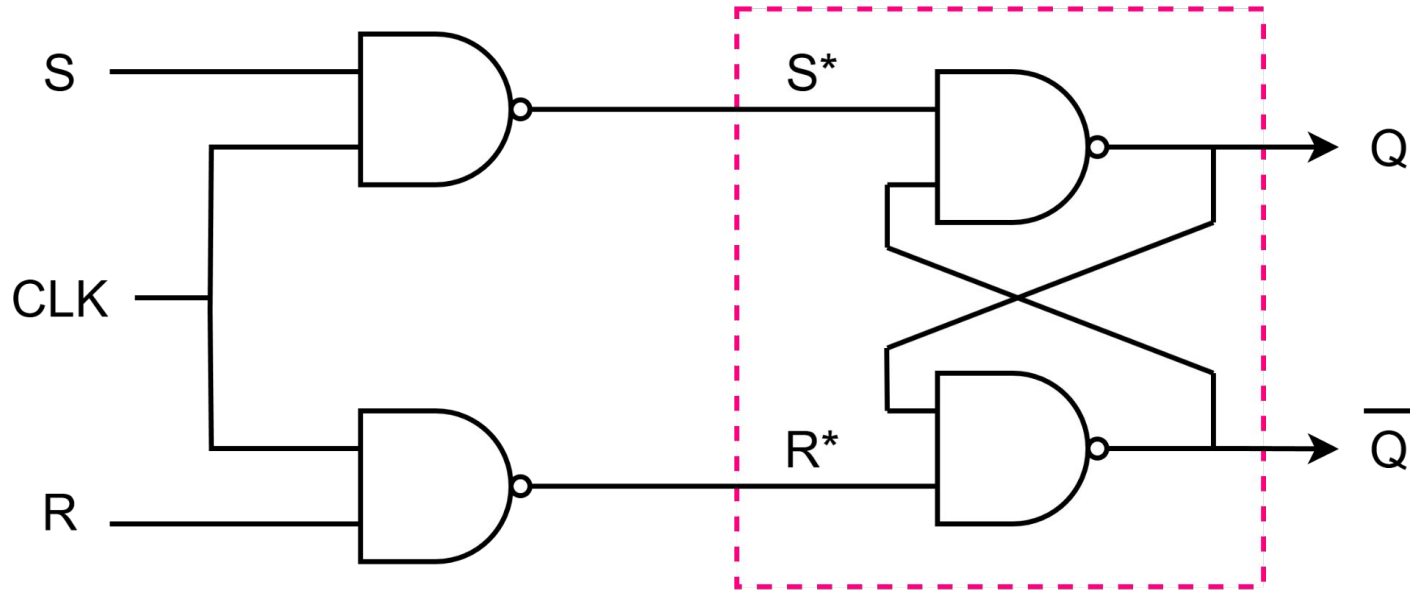
CASE	S	R	Q	Q'	Notes
1	0	1	1	0	Calculation of Q done first
	1	1	1	0	Q and Q' remains unchanged
2	1	0	0	1	Calculation of Q' done first
	1	1	0	1	Q and Q' remains unchanged
3	0	0	1	1	Impossible!

SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/ No Change	



SR Flip Flop



Now we will use the knowledge of SR latch to build the truth table for SR FF. But before that, we need to do some calculation.

SR Flip Flop

$$S^* = (S.CLK)' = S' + CLK'$$

$$R^* = (R.CLK)' = R' + CLK'$$

Now, for a FF to work,
CLK must be 1. If CLK=1, then,

$$S^* = S' + 0 = S'$$

$$R^* = R' + 0 = R'$$

If,

S=0, R=0, then $S^*=1$, $R^*=1$

S=0, R=1, then $S^*=1$, $R^*=0$

S=1, R=0, then $S^*=0$, $R^*=1$

S=1, R=1, then $S^*=0$, $R^*=0$

Let's build the truth table now [this table also includes latch inputs, which are S^* and R^*].

S	R	S^*	R^*	Q	Q'
0	0	1	1	Memory/ No Change	
0	1	1	0	0	1
1	0	0	1	1	0
1	1	0	0	Not Used	

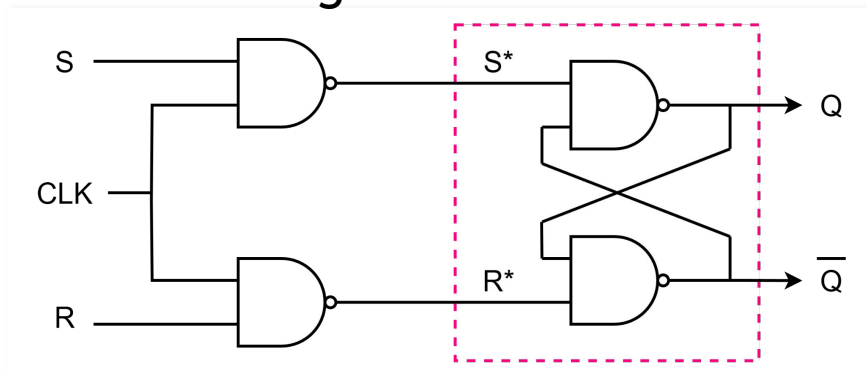
SR FF Final Truth Table

S	R	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Not Used	

Clocked S-R FF Diagram

S-R FF + Clock Pulse (CP) and 2 NAND gates → Clocked *S-R FF*.

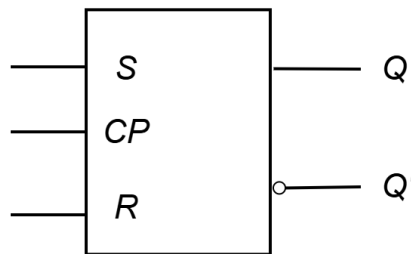
Circuit Diagram



Truth Table

CLK	S	R	Q	Q'
0	X	X	X	
1	0	0	Memory/ No Change	
1	0	1	0	1
1	1	0	1	0
1	1	1	Not Used	

Block Diagram



Types of tables in sequential circuit

- ❖ Truth/Criteria Table
 - Lists all possible input combinations and corresponding outputs. [FF input = FF output]
- ❖ Characteristic Table
 - Shows the flip-flop's next state for given inputs and current state. [Present State + FF input = Next State]
- ❖ Excitation Table
 - Shows the required inputs to flip-flops to cause a desired state transition. [Present State + Next State = FF input]

S-R Flip-Flop Truth + Characteristics Table

S	R	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Not Used	

While building the Characteristics table, always consider Q from the truth table as Q' is just the complement of Q.

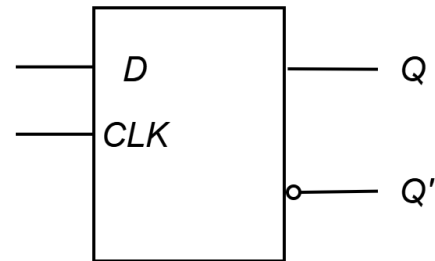
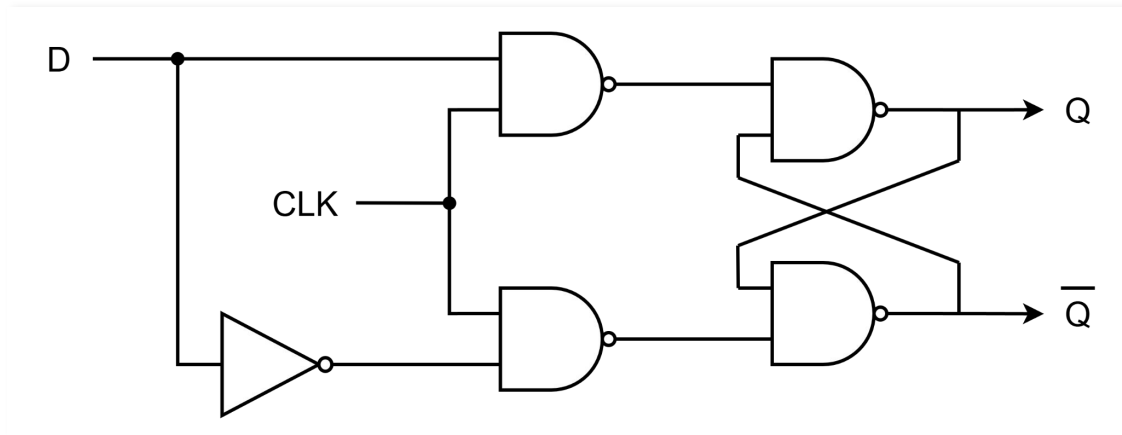
Characteristic Table

Q(t)	S	R	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	indeterminate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	indeterminate

You always need to remember the truth table to build the characteristics table. Meaning, truth table comes first, characteristics table comes second. You can't build characteristics table before truth table

Clocked D- Flip Flop

- ❖ Make R input equal to $S' \rightarrow D$ FF
- ❖ D FF eliminates the undesirable condition of invalid state in the S-R FF.





D - FF Truth Table + Characteristic Table

D - FF Truth Table

D	Q	Q'
0	0	1
1	1	0

D - FF Characteristic Table

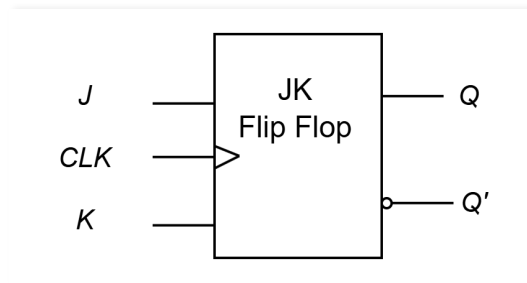
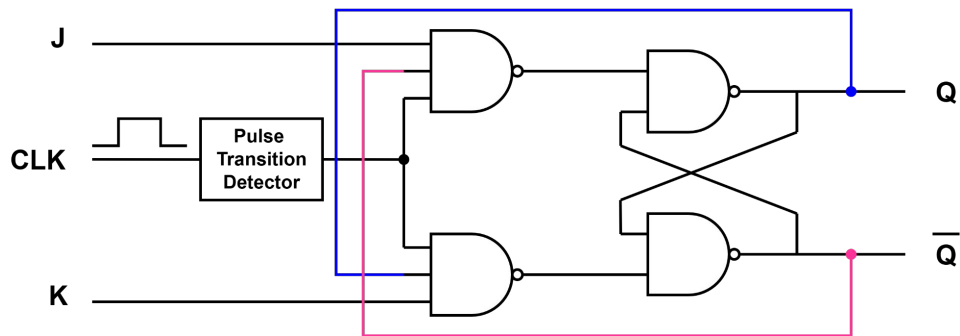
$Q(t)$	D	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1



J-K Flip Flop

- ❖ **J-K flip-flop:** Q and Q' are fed back to the NAND gates.
- ❖ **No invalid state.**
- ❖ **Include a toggle state.**
 - J=HIGH (and K=LOW) ☐ SET state [Q = 1]
 - K=HIGH (and J=LOW) ☐ RESET state [Q = 0]
 - both inputs LOW ☐ no change [Memory]
 - both inputs HIGH ☐ toggle [Q = Q']

J-K Flip Flop



JK
Flip Flop
Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

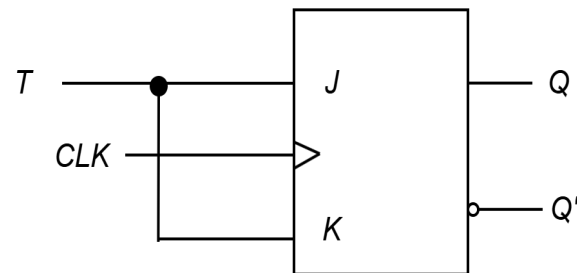
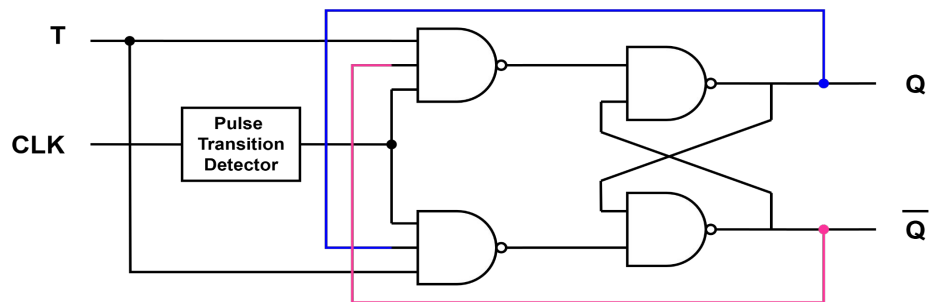
Characteristic
Table

Q(t)	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



T Flip Flop

- ❖ **T flip-flop** : single-input version of the J-K flip-flop, formed by tying both inputs together.



T - FF Truth Table

T	Q	Q'
0	No change	
1	Toggle	

Characteristic Table

$Q(t)$	T	$Q(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0



Try it Yourself (Self Study)

(Learn it yourself) (Draw both circuit and block diagram)

- ❖ Design a D FF using SR FF
- ❖ Design a T FF using JK FF
- ❖ Design a D FF using JK FF

Flip Flop Excitation Tables

- ❖ **Excitation Tables** : Given the required transition from present state to next state, determine the flip-flop input(s)

JK Flip-Flop

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

SR Flip-Flop

Q	Q^+	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	X	0

T Flip-Flop

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

D Flip-Flop

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

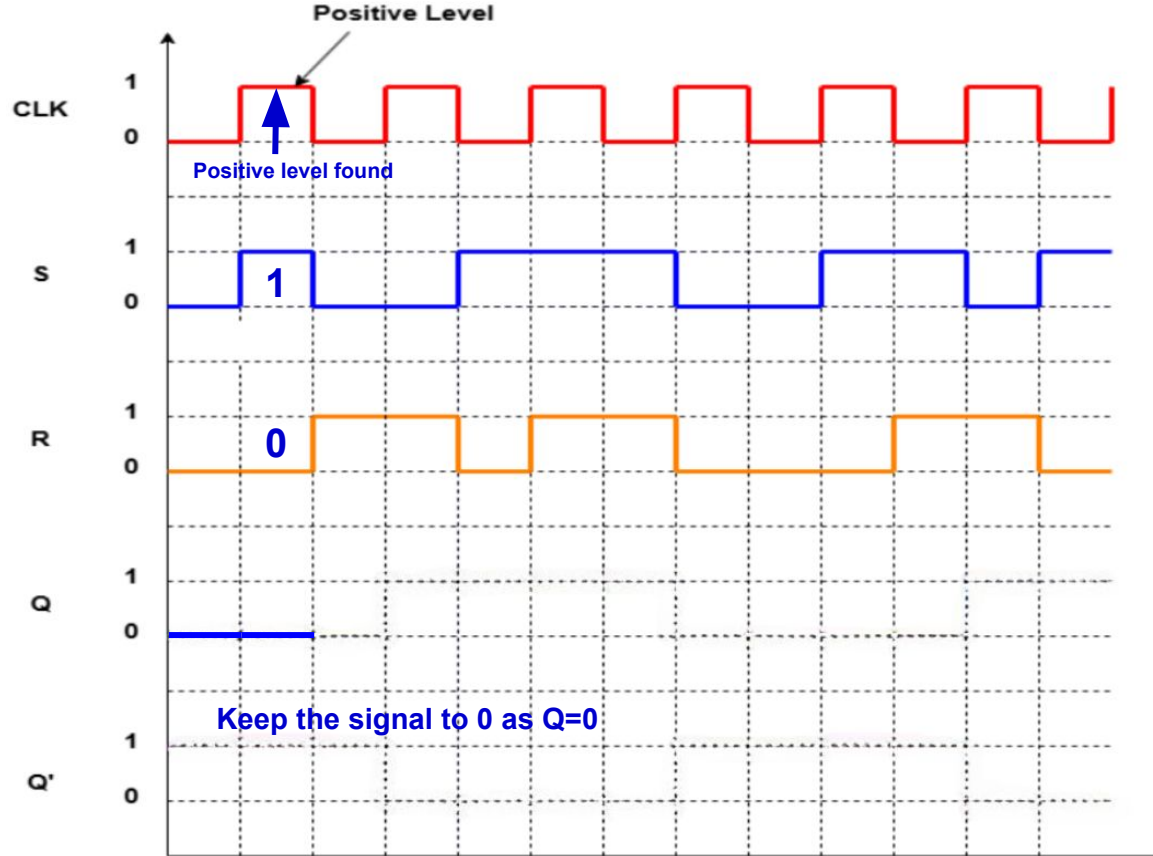
Analyzing Flip-Flop Behavior with Timing Diagrams

❖ 4 types based on the triggering mechanism of the Flip Flop

- Positive Level Triggered
 - Negative Level Triggered
- } Latch

- Positive/ Rising Edge Triggered
 - Negative/ Falling Edge Triggered
- } Flip Flop

Positive Level Triggered Timing Diagram



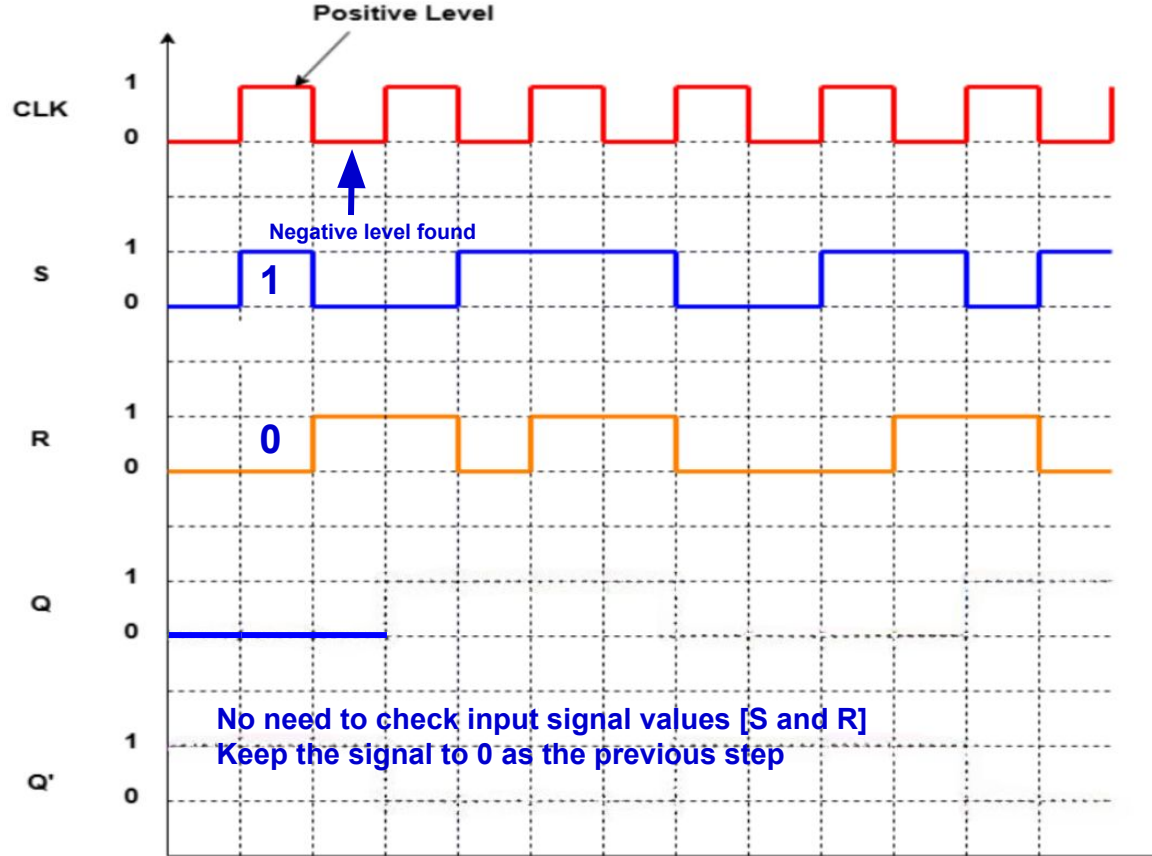
CLK, S, R signal will be
given in question
Positive - 1
Negative - 0

SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

Truth table will not be given in the
question

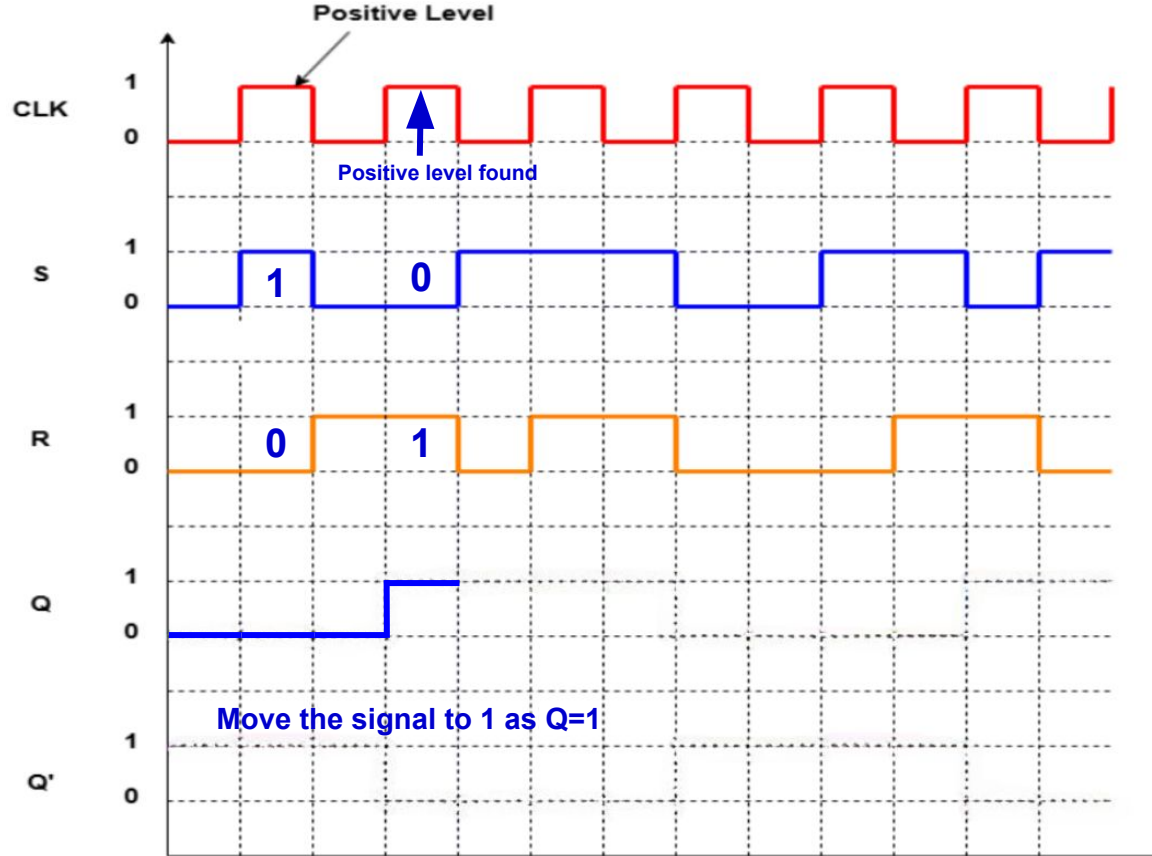
Positive Level Triggered Timing Diagram



SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

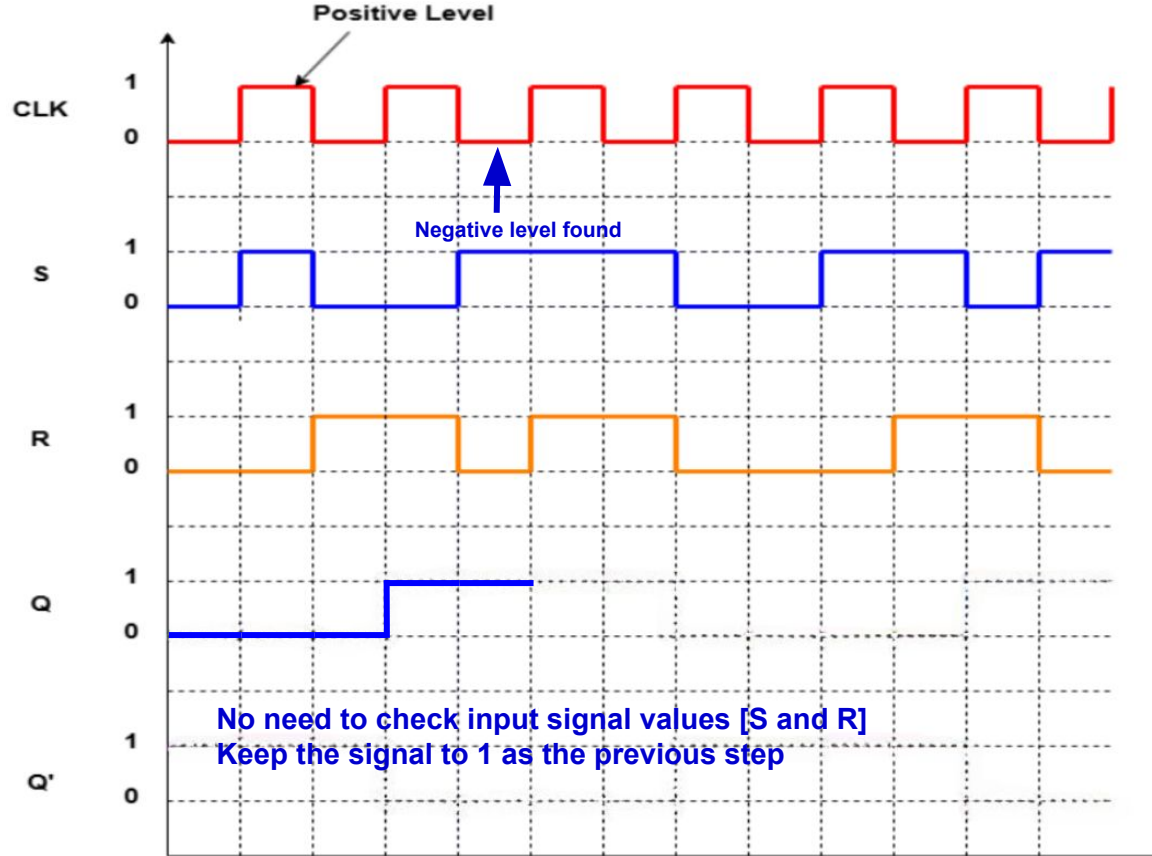
Positive Level Triggered Timing Diagram



SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

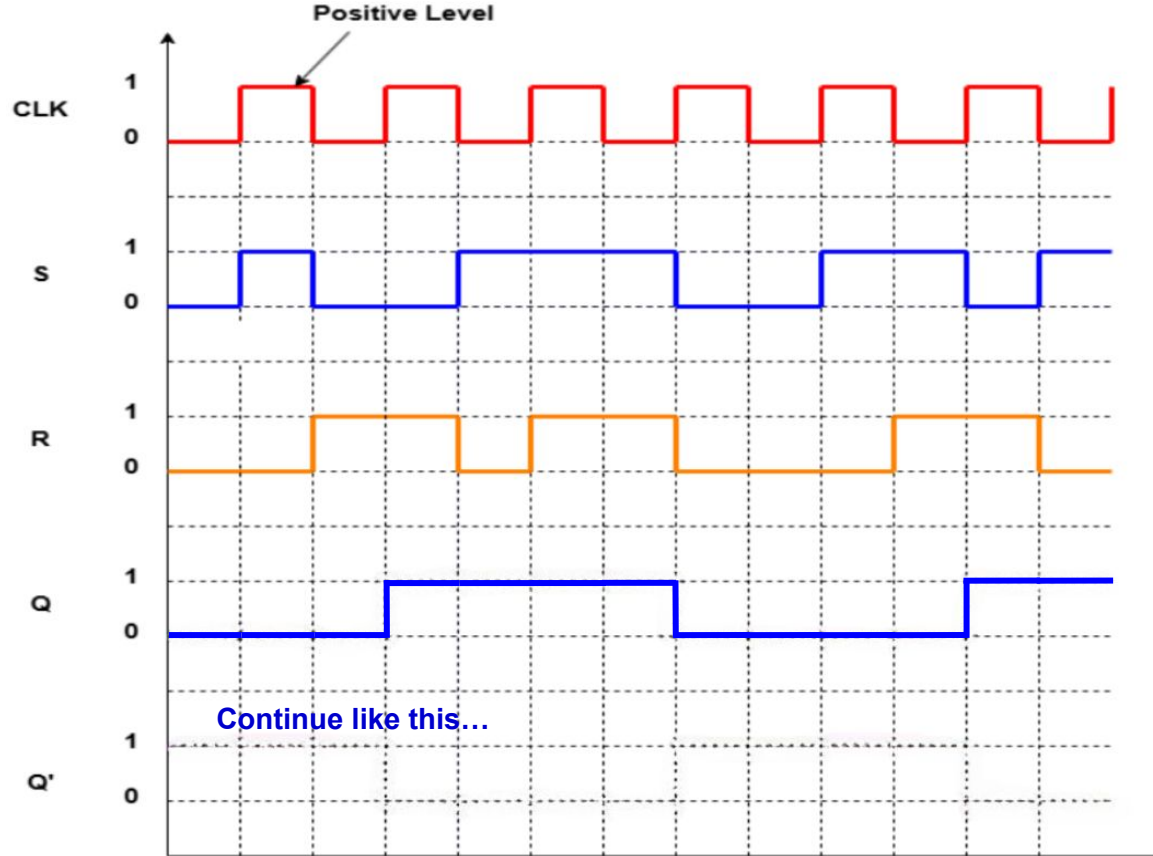
Positive Level Triggered Timing Diagram



SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

Positive Level Triggered Timing Diagram

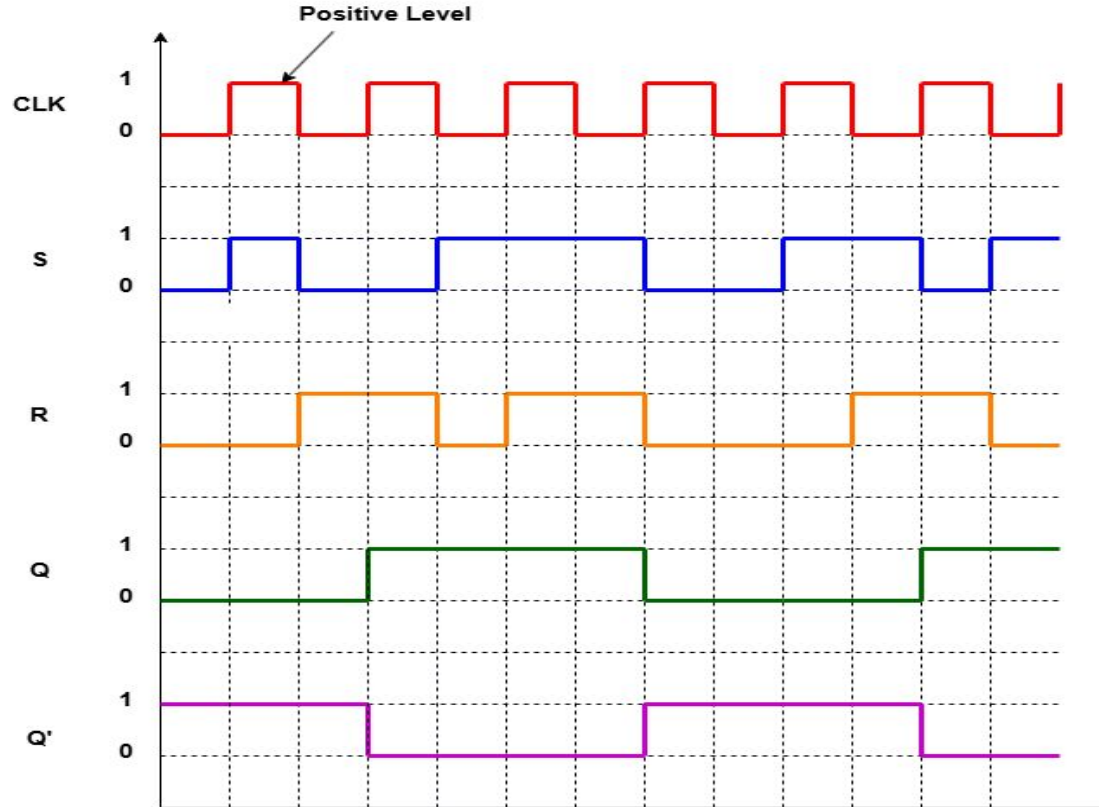


SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

Positive Level Triggered Timing Diagram

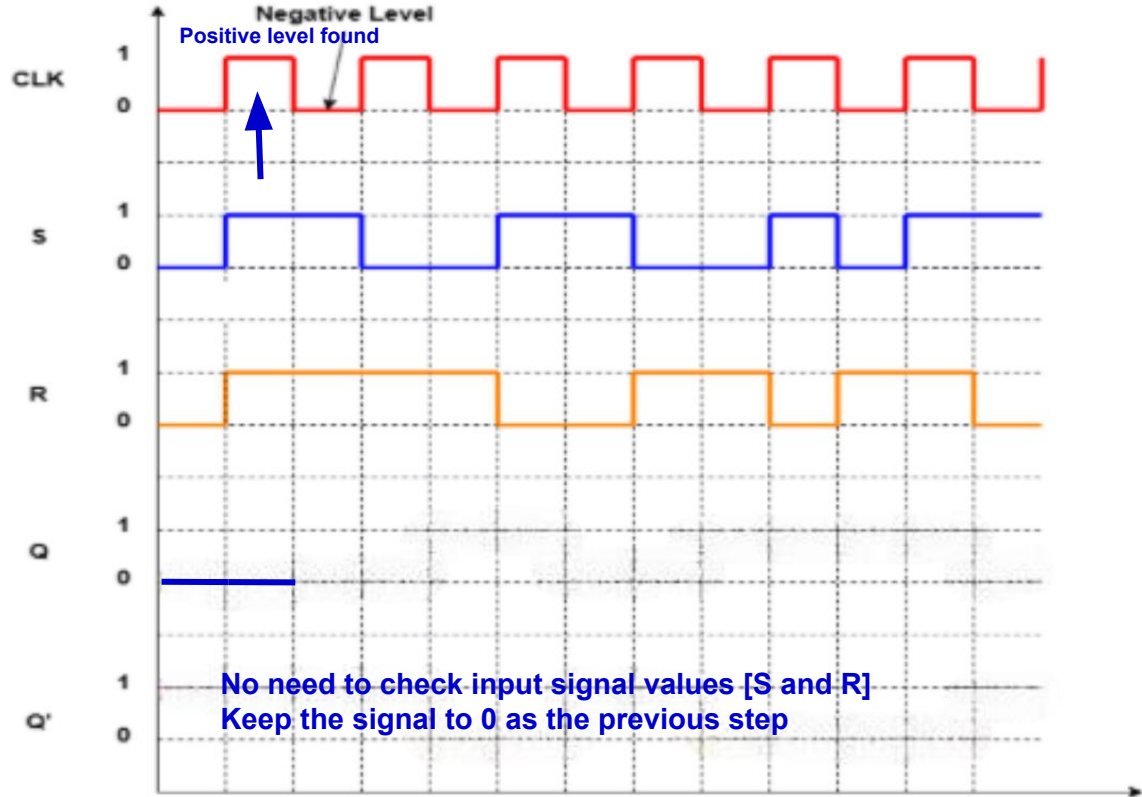
Assume $Q = 0$, whenever invalid condition occurs



SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

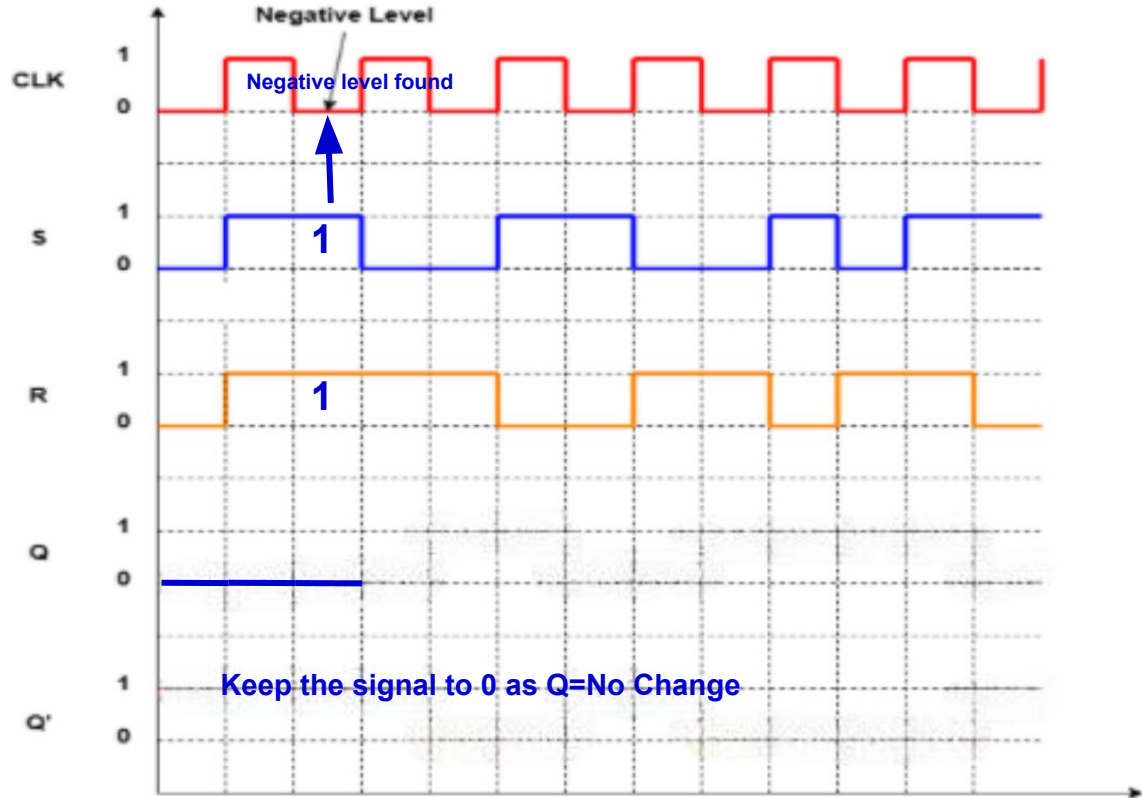
Negative Level Triggered Timing Diagram



SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

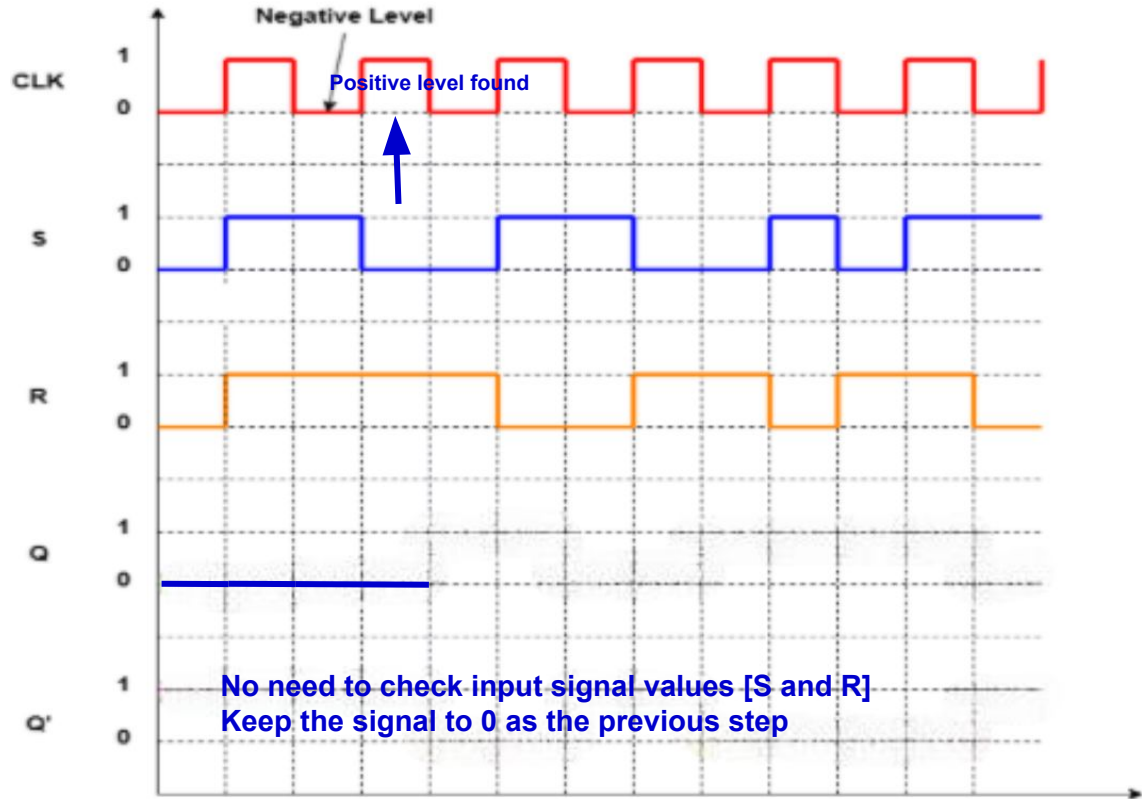
Negative Level Triggered Timing Diagram



SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

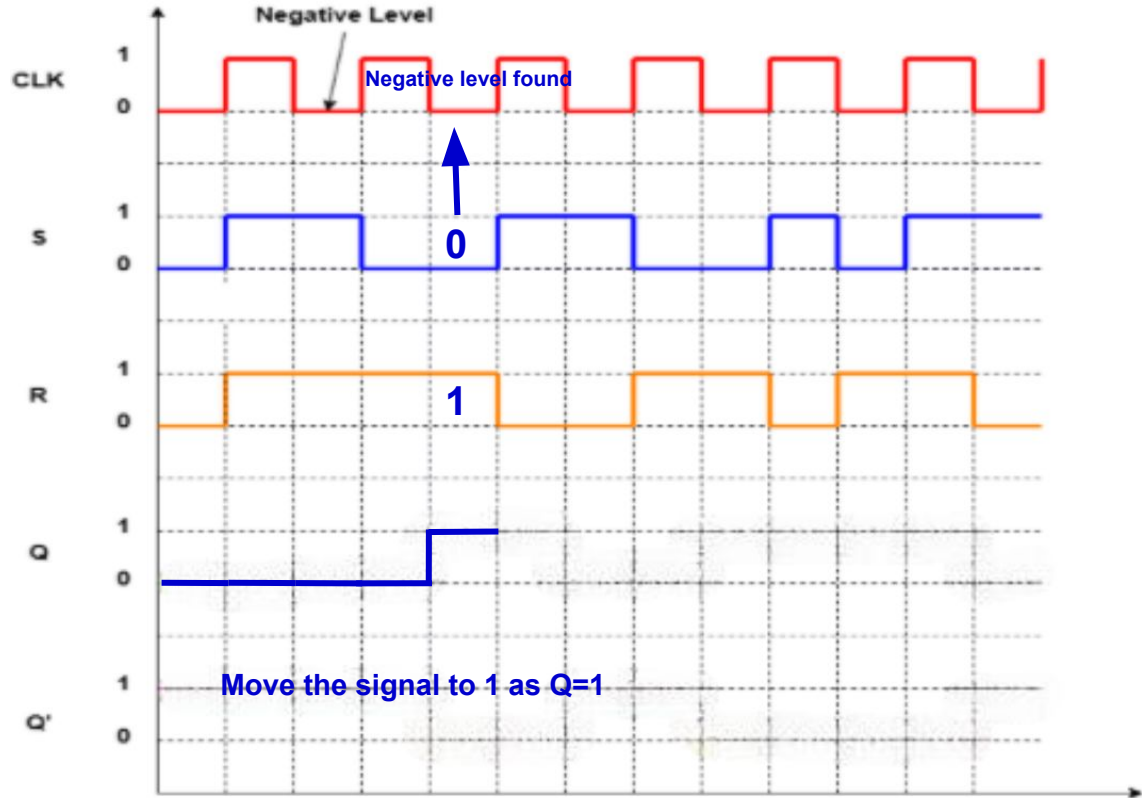
Negative Level Triggered Timing Diagram



SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

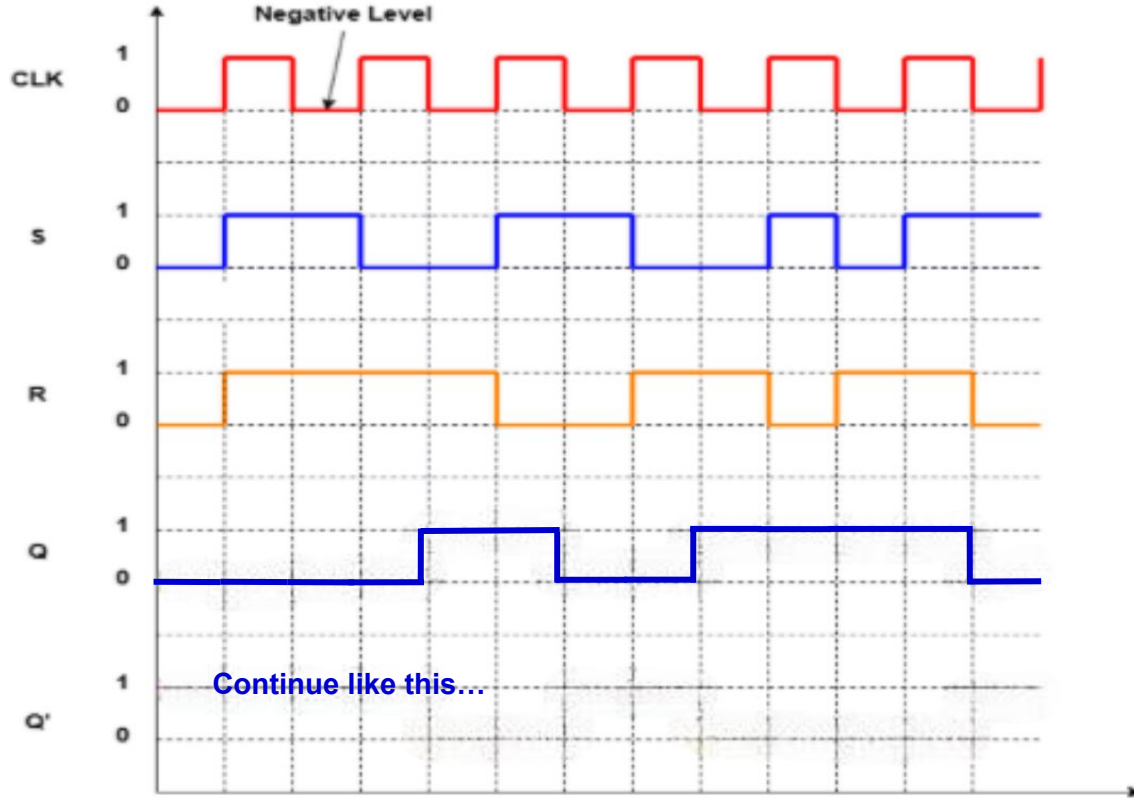
Negative Level Triggered Timing Diagram



SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

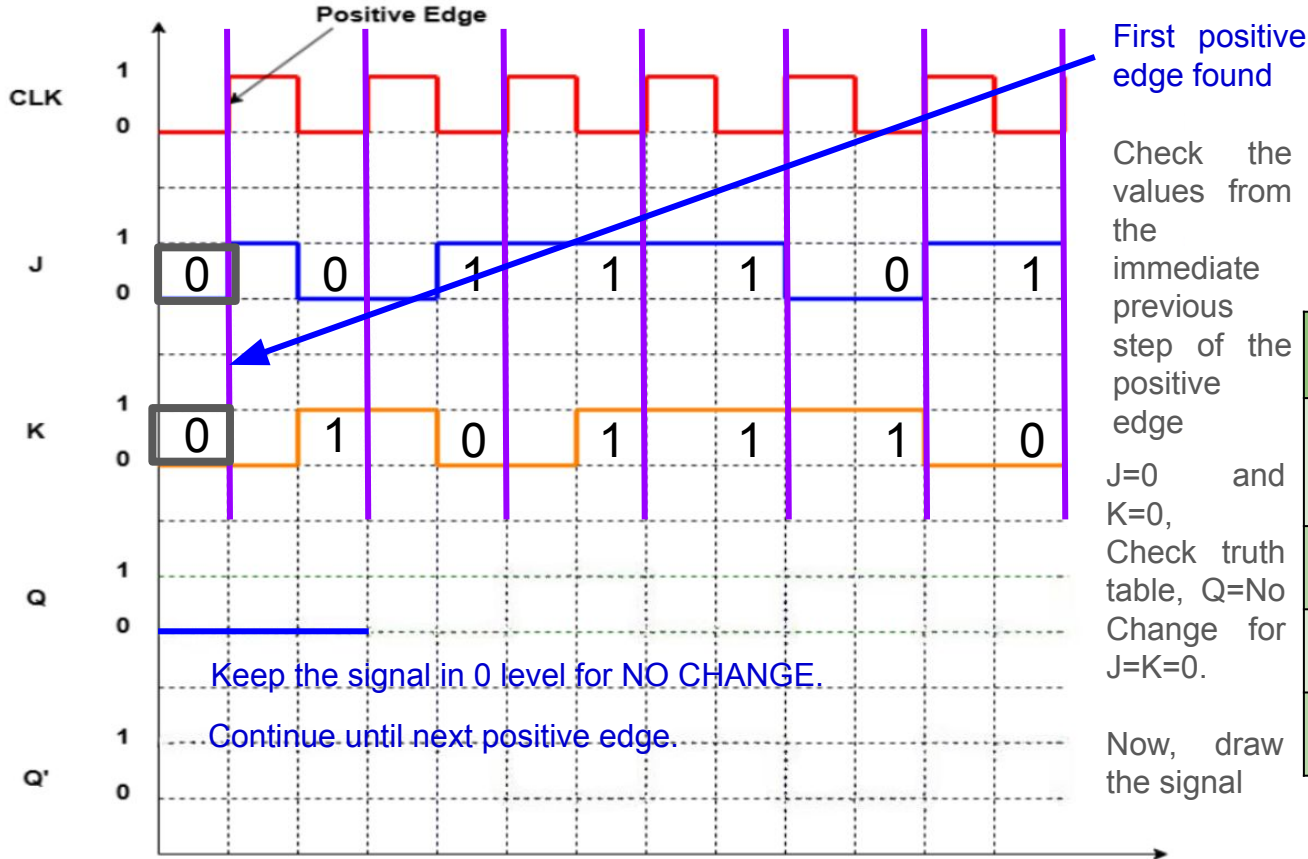
Negative Level Triggered Timing Diagram



SR Latch Truth Table

S	R	Q	Q'
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	Memory/No Change	

Positive Edge Triggered Timing Diagram



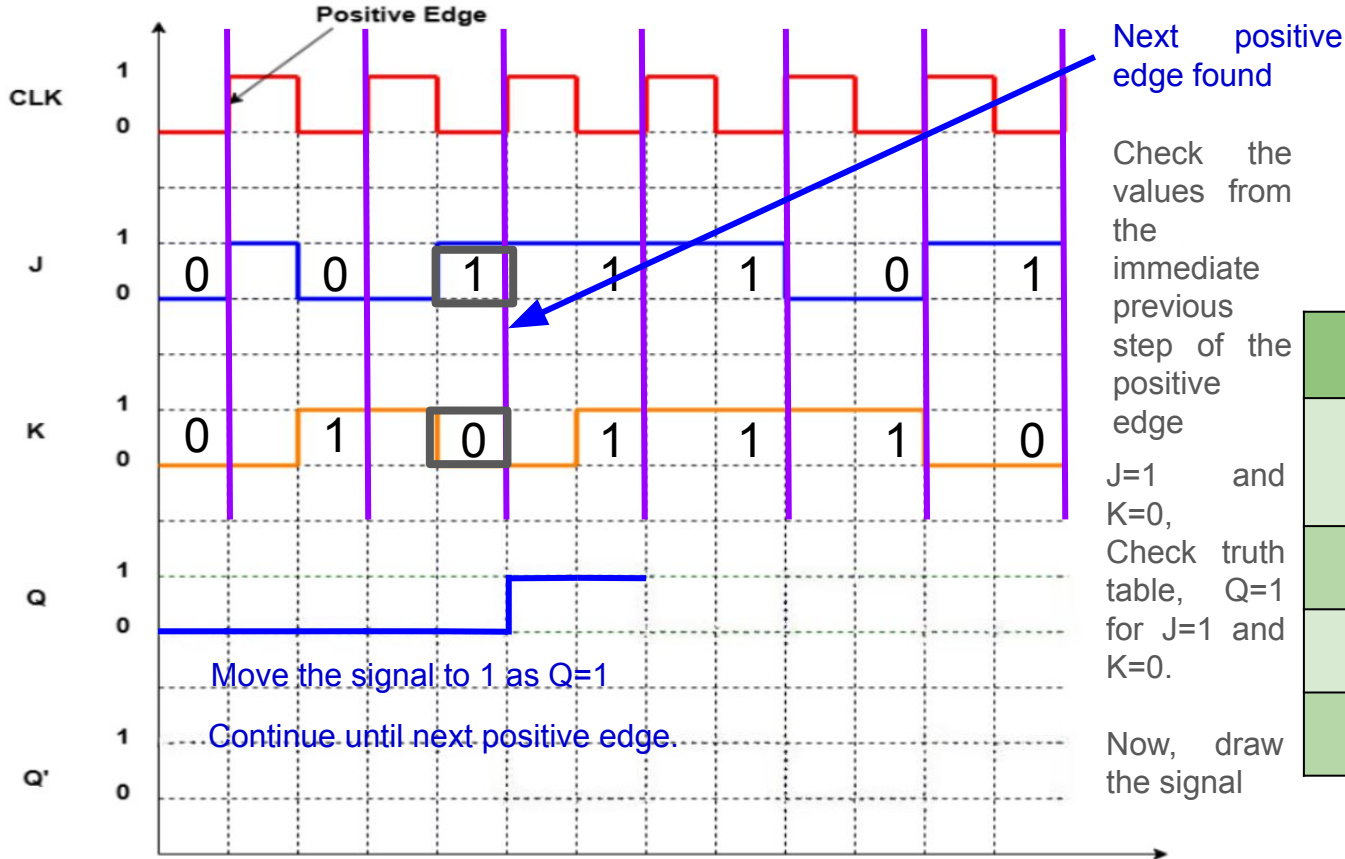
JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	



J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

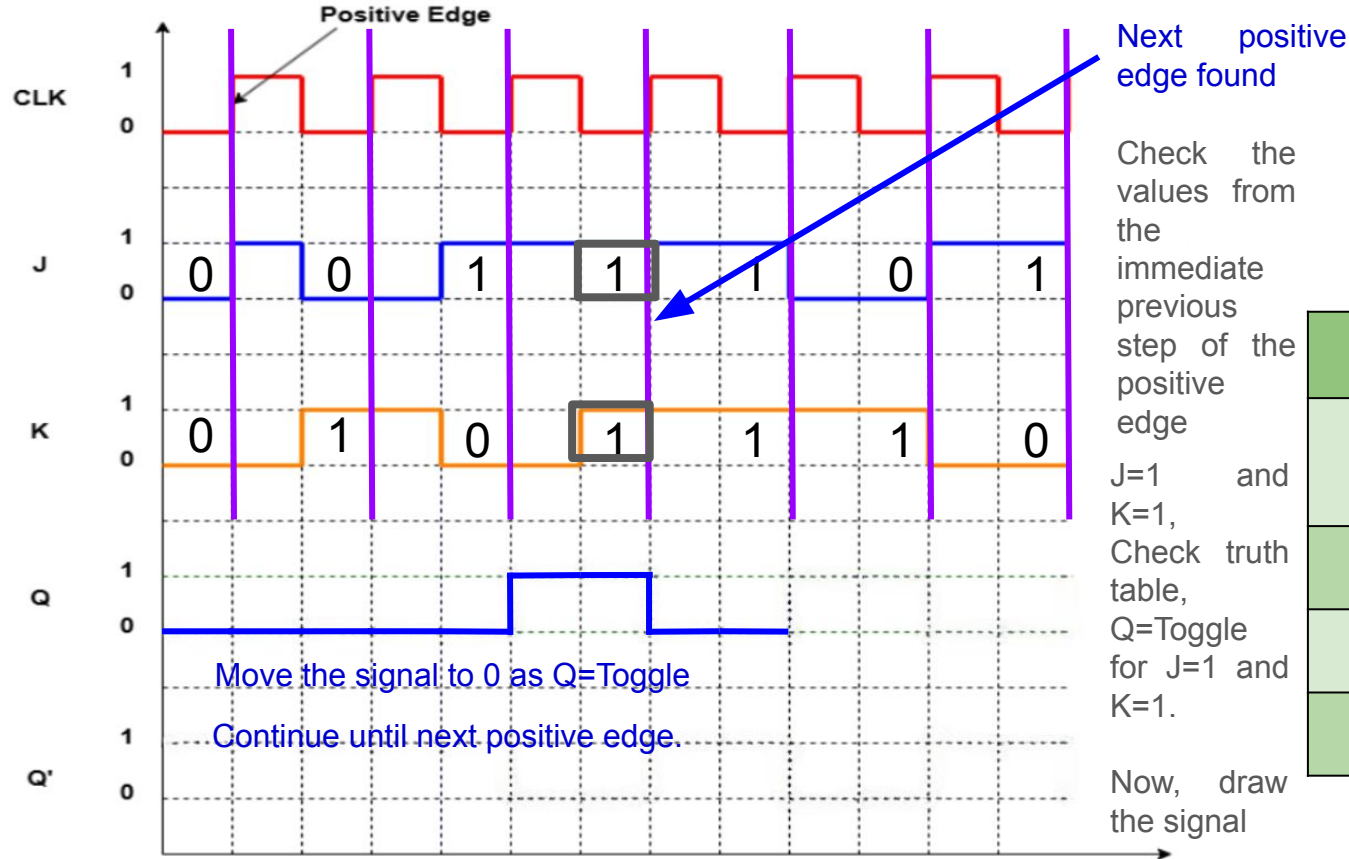
Positive Edge Triggered Timing Diagram



JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

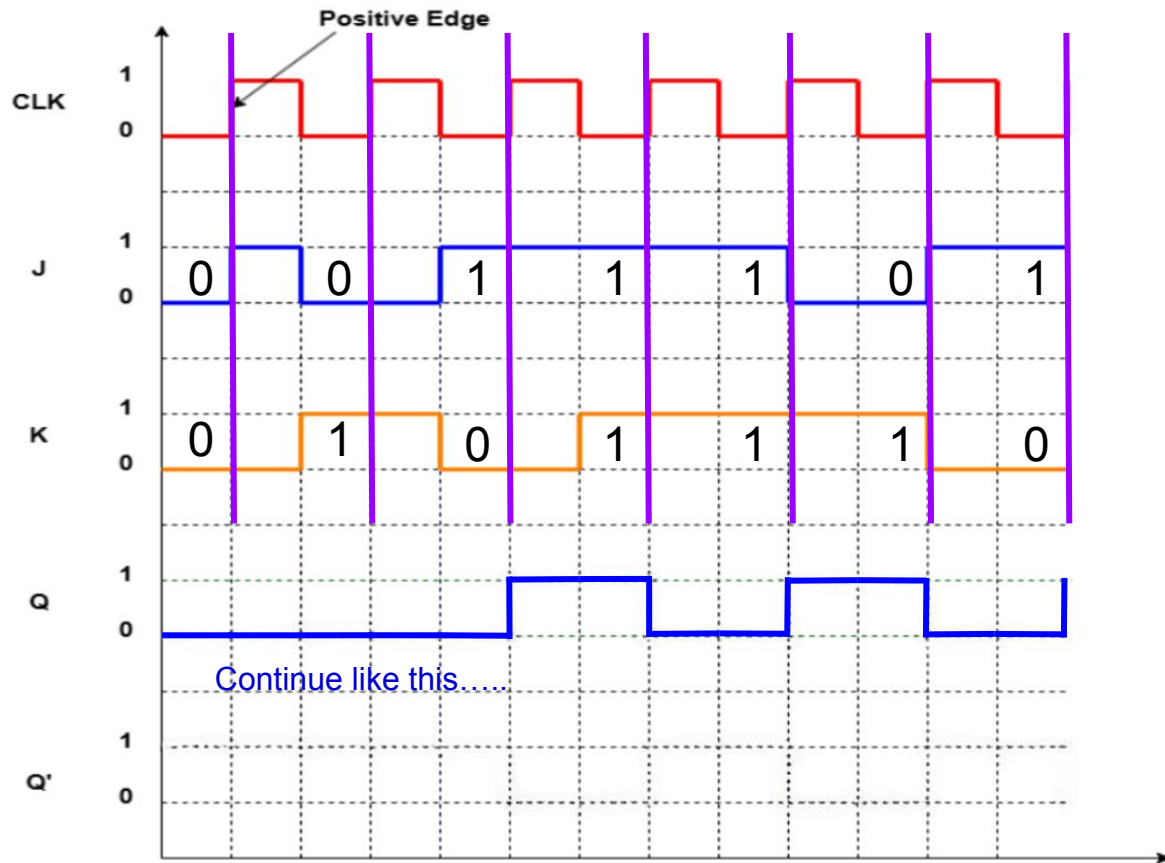
Positive Edge Triggered Timing Diagram



JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

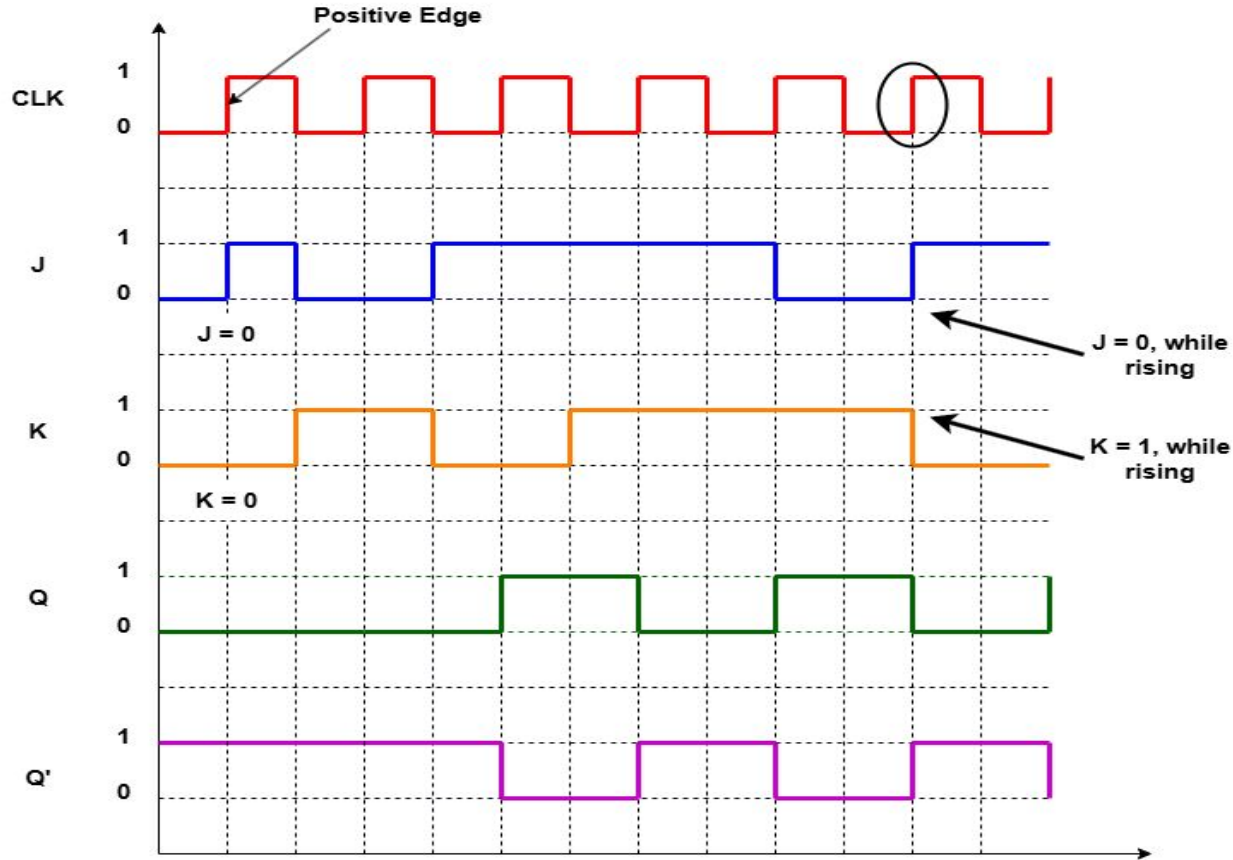
Positive Edge Triggered Timing Diagram



JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

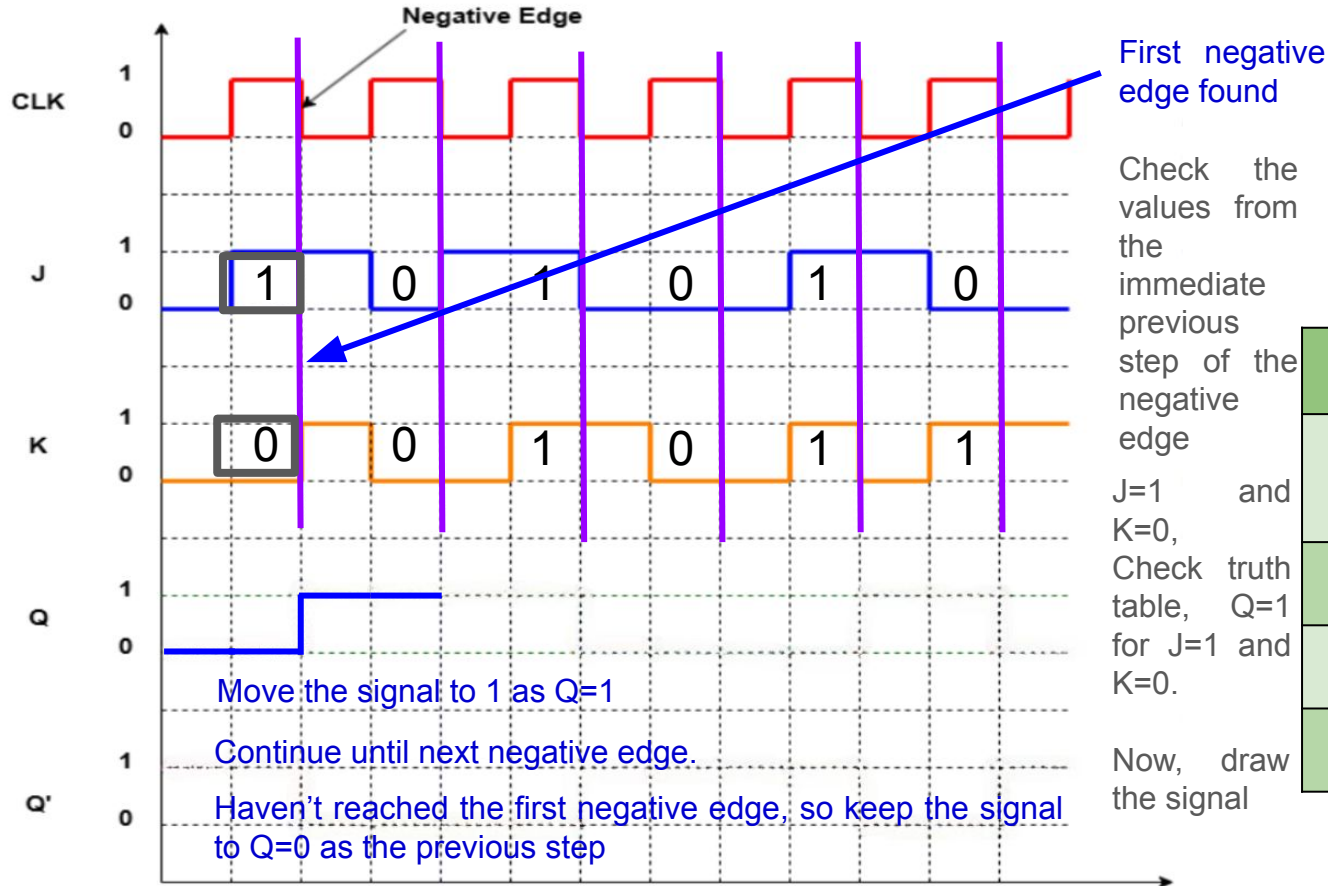
Positive Edge Triggered Timing Diagram



JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

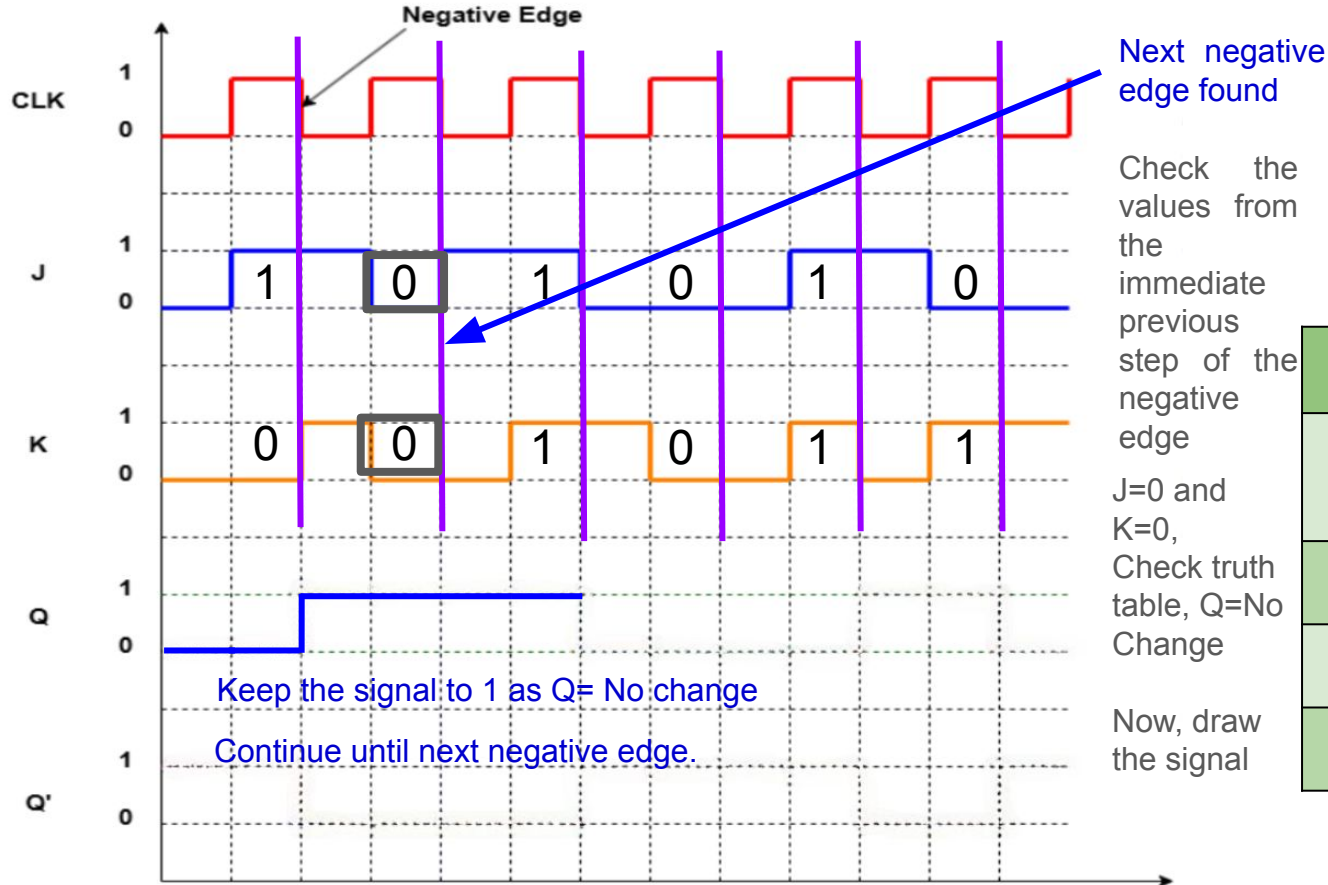
Negative Edge Triggered Timing Diagram



JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

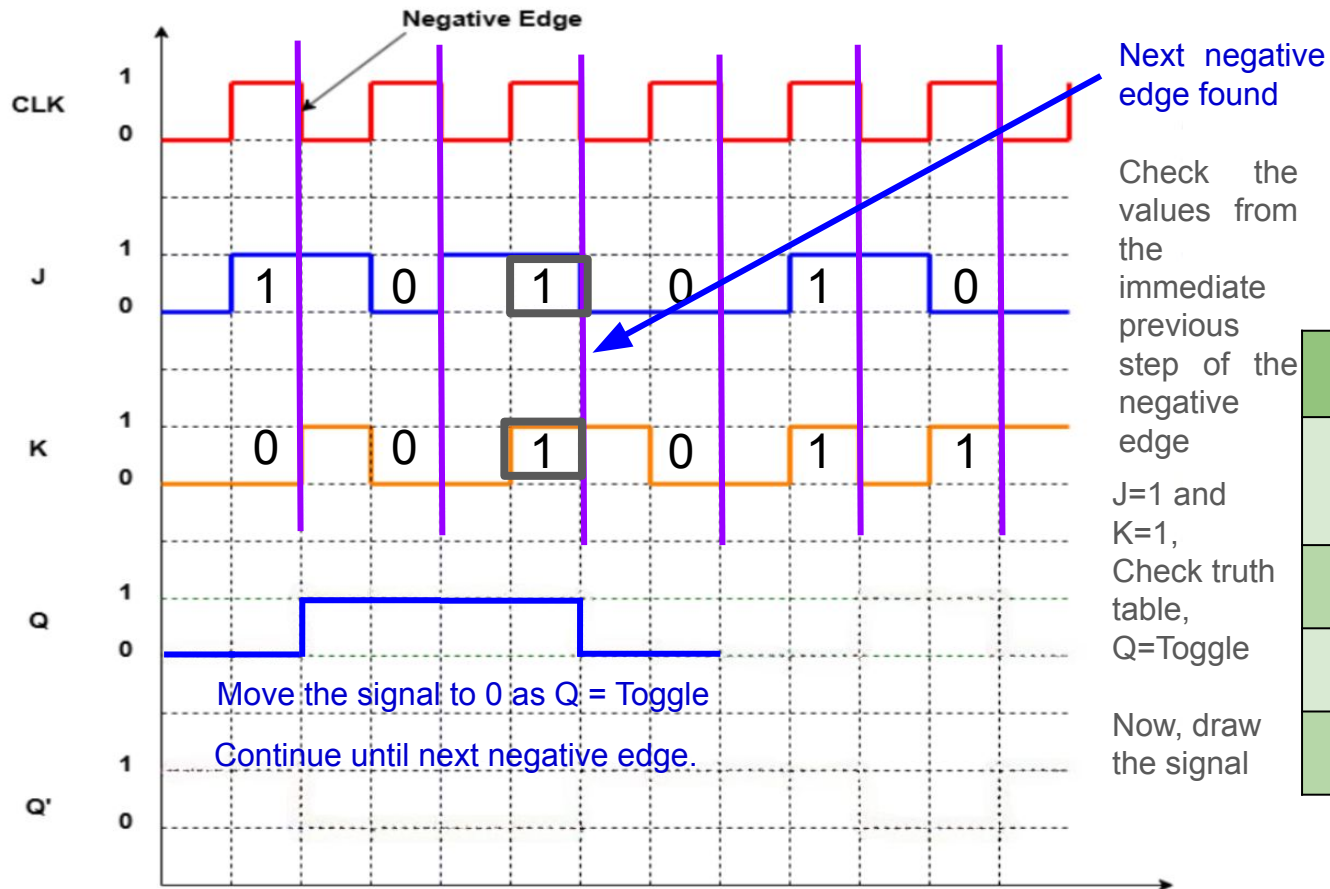
Negative Edge Triggered Timing Diagram



JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

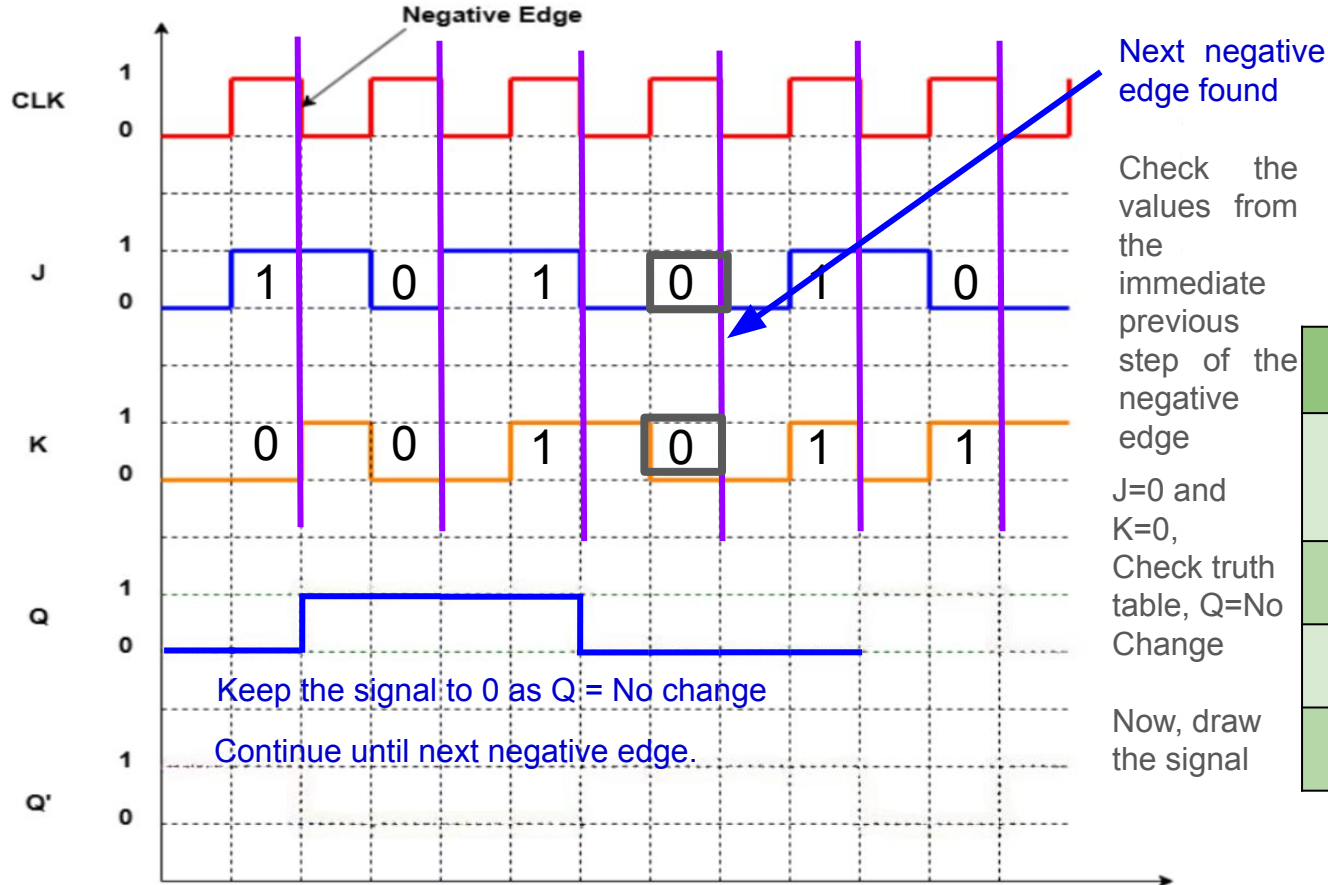
Negative Edge Triggered Timing Diagram



JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

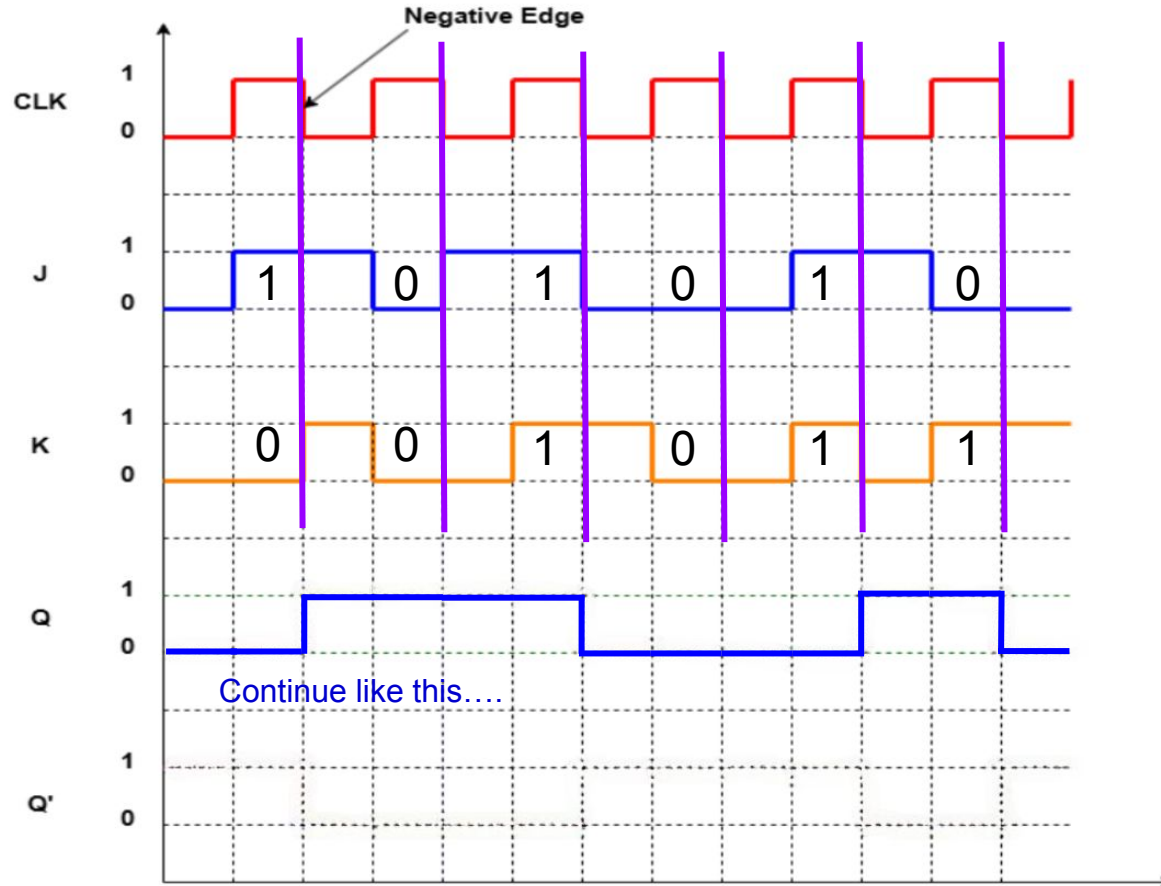
Negative Edge Triggered Timing Diagram



JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

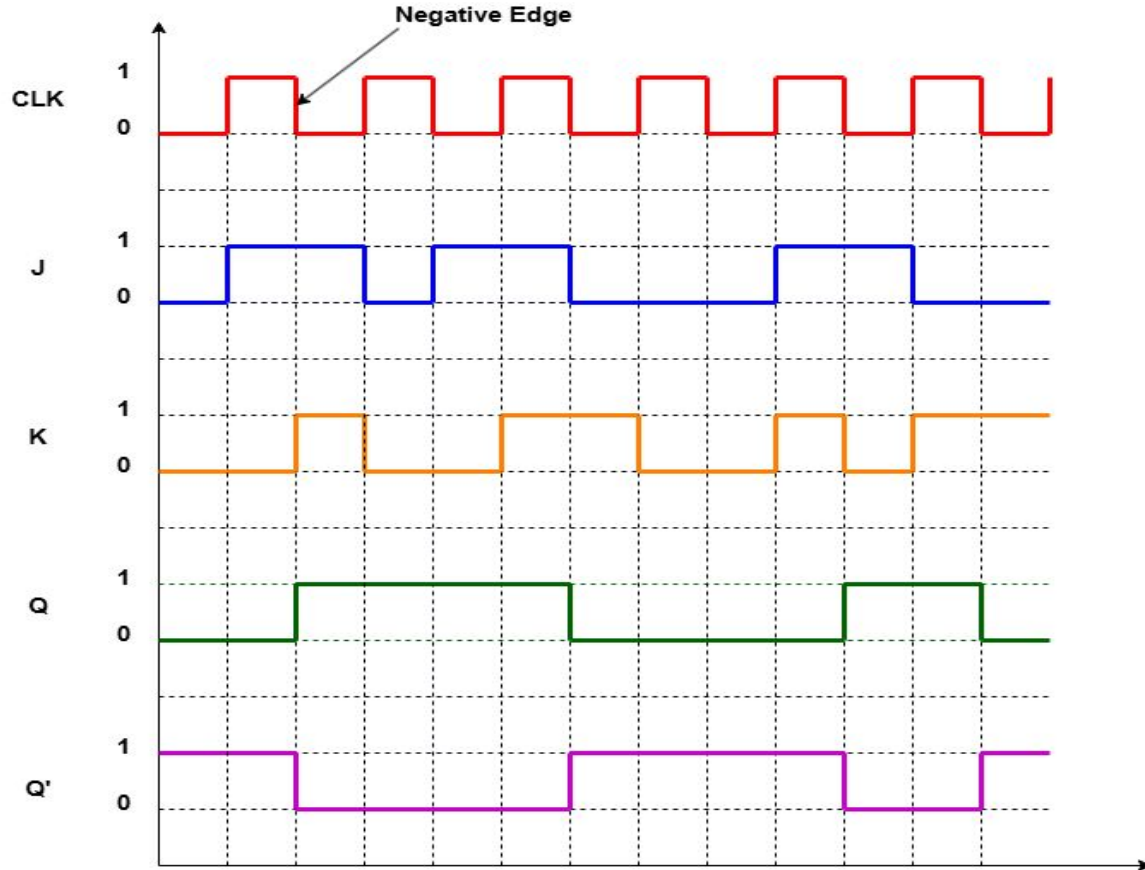
Negative Edge Triggered Timing Diagram



JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

Negative Edge Triggered Timing Diagram

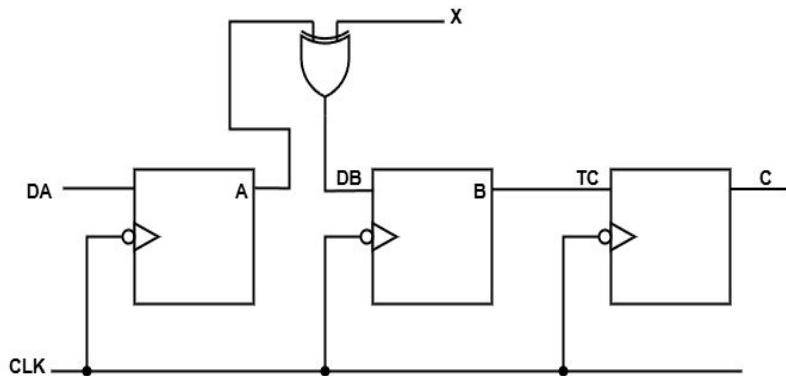


JK FF Truth Table

J	K	Q	Q'
0	0	Memory/ No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	



Solution (CLK, DA and X are Given)



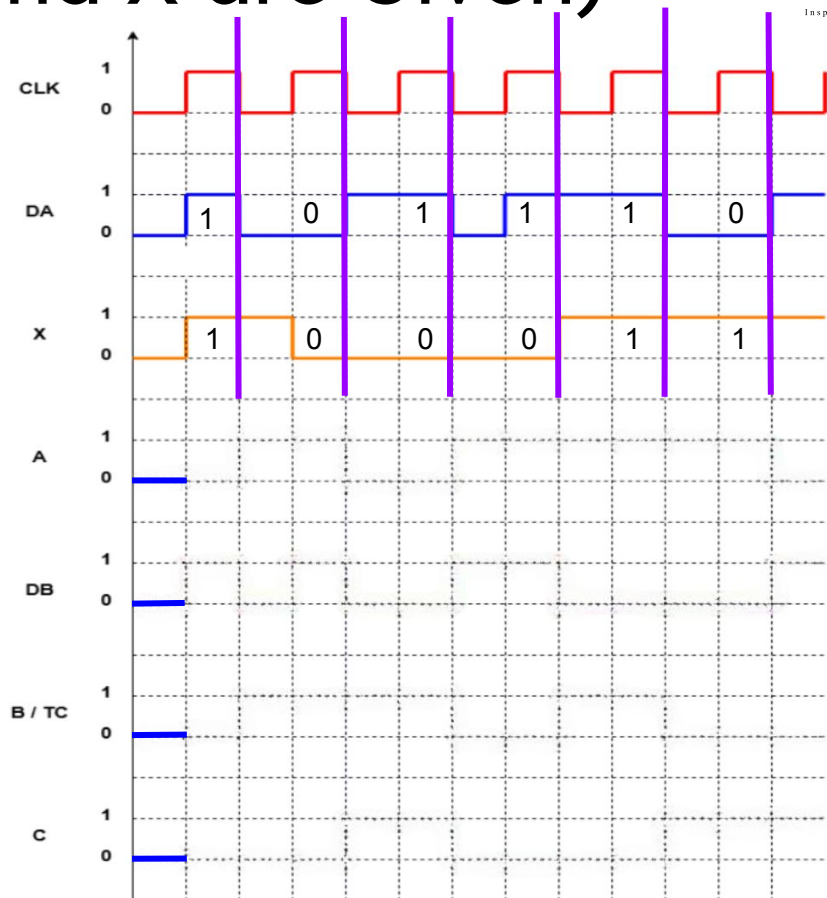
Here,
Find out value of A based upon D FF's truth table
DB = A xor X [use XOR gate truth table to find out values]
Find out value of C based upon T FF's truth table

T - FF Truth Table

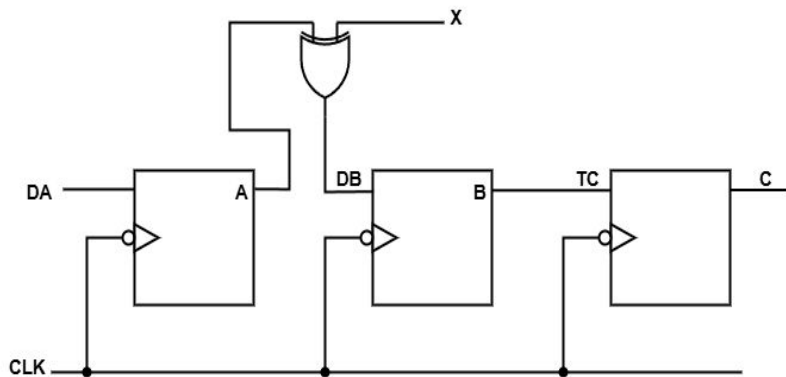
T	Q	Q'
0	No change	
1	Toggle	

D - FF Truth Table

D	Q	Q'
0	0	1
1	1	0



Solution (CLK, DA and X are Given)



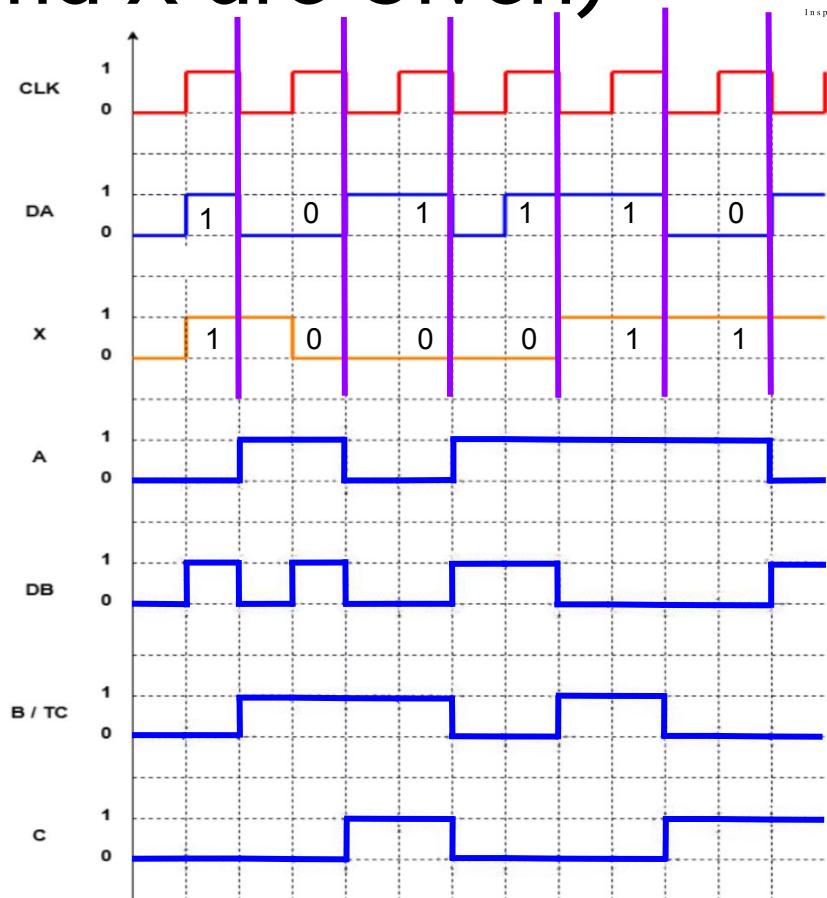
Here,
Find out value of A based upon D FF's truth table
DB = A xor X [use XOR gate truth table to find out values]
Find out value of C based upon T FF's truth table

T - FF Truth Table

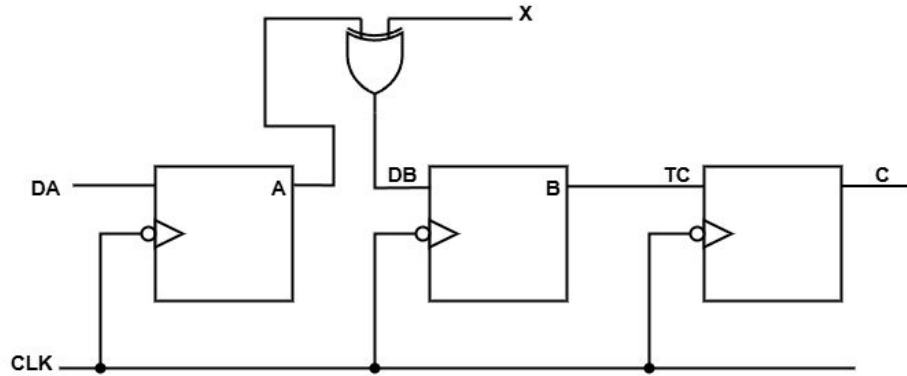
T	Q	Q'
0	No change	
1	Toggle	

D - FF Truth Table

D	Q	Q'
0	0	1
1	1	0



Solution (CLK, DA and X are Given)



Here,
Find out value of A based upon D FF's truth table
DB = A xor X [use XOR gate truth table to find out values]
Find out value of C based upon T FF's truth table

T - FF Truth Table

T	Q	Q'
0	No change	
1	Toggle	

D - FF Truth Table

D	Q	Q'
0	0	1
1	1	0

